

Master's Programme in Information Technology

Systematic Mapping and Benchmarking of Sign Language Datasets

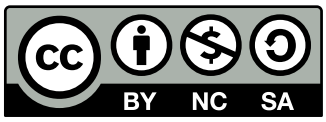
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Abstract

Sign languages serve as the primary means of communication for millions of individuals who are deaf or hard of hearing. Developing computational tools that can automatically recognize and translate signed communication depends directly on the availability of high-quality datasets. These datasets are used to train and evaluate machine learning models across tasks such as isolated sign language recognition, continuous sign language recognition, and sign language translation. Existing surveys have largely treated datasets as supporting material for evaluating models rather than as the primary object of structured analysis, leaving researchers without a consolidated reference for selecting datasets by task, language, or modality. This thesis applies an established systematic mapping methodology, supplemented by systematic literature review guidelines, to produce a structured classification of publicly available sign language datasets. Thirty datasets were identified through a keyword-based search across five academic databases, supplemented by backward snowballing, and analyzed using a predefined extraction scheme of 38 attributes across five thematic attribute groups. The mapped corpus encompasses 23 distinct sign languages across five geographic regions. RGB video is the dominant modality, present in 29 of the 30 datasets, while depth video and multi-view configurations are available in only 7 and 5 datasets, respectively. Isolated sign language recognition is the most strongly supported task, while sentence-level translations are present in only 15 datasets and non-manual feature annotation in only 3. Geographic coverage is uneven, with limited representation from Africa, Latin America, and large parts of Asia. The data extraction sheets produced in this study are publicly available to support replication. The study identifies six recurring structural gaps and provides targeted recommendations alongside a task-based dataset selection guide to assist researchers in choosing suitable datasets for specific computational tasks.

Keywords Sign Language Dataset, Sign Language Recognition, Sign Language Translation, Isolated Sign Language, Continuous Sign Language, Mixed Sign Language

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Abbreviations and Terms

Abbreviations

ACM	Association for Computing Machinery
BLEU	Bilingual Evaluation Understudy
CEFR	Common European Framework of Reference for Languages
CODA	Children of Deaf Adults
CSLR	Continuous Sign Language Recognition
ELAN	EUDICO Linguistic Annotator
FATE	Fairness, Accountability, Transparency, and Ethics
FPS	Frames Per Second
GRU	Gated Recurrent Unit
I3D	Inflated 3D Convolutional
IEEE	Institute of Electrical and Electronics Engineers
ISLR	Isolated Sign Language Recognition
L1	First Language
L2	Second Language
RGB	Red, Green, Blue
RGB-D	Red, Green, Blue-Depth
SLR	Sign Language Recognition
SLT	Sign Language Translation
SVM	Support Vector Machine
WER	Word Error Rate

Sign Languages

ASL	American Sign Language
Auslan	Australian Sign Language
AzSL	Azerbaijani Sign Language
BdSL	Bangla Sign Language
BSL	British Sign Language
CSL	Chinese Sign Language
DGS	German Sign Language
DSGS	Swiss German Sign Language
FinSL	Finnish Sign Language
GSL	Greek Sign Language
ISL	Indian Sign Language
KSL	Korean Sign Language
LIS	Italian Sign Language
LSA	Argentinian Sign Language
LSE	Spanish Sign Language
LSF	French Sign Language

LSR	Romanian Sign Language
RSL	Russian Sign Language
SL	Sign Language
SSL	Saudi Sign Language
TSL	Turkish Sign Language

Datasets

Alabib-65	Algerian Sign Language Dataset
AUTSL	Ankara University Turkish Sign Language Dataset
AzSLD	Azerbaijani Sign Language Dataset
BdSLW60	Bangla Sign Language Word-Level 60
BOBSL	BBC-Oxford British Sign Language Dataset
BosphorusSign22k	Bosphorus Sign Language Dataset
CASL-W60	Central African Sign Language Word-Level 60
CE-CNSL	Complex Environment Chinese Sign Language Dataset
Corpus FinSL	Corpus of Finnish Sign Language
CSL-Daily	Chinese Sign Language Daily Dataset
How2Sign	How2Sign American Sign Language Dataset
INCLUDE	Indian Lexicon Sign Language Dataset
KSL-Guide	Korean Sign Language Guidance Dataset
KSU-SSL	King Saud University Saudi Sign Language Database
LIBRAS-UFOP	Brazilian Sign Language Dataset
LSA64	Argentinian Sign Language Dataset
Mediapi-RGB	French Sign Language RGB Video Dataset
MM-WLAuslan	Multi-View Multi-Modal Word-Level Australian Sign Language Dataset
MS-ASL	Microsoft American Sign Language Dataset
OpenASL	Open American Sign Language Dataset
RoCoISLR	Romanian Continuous and Isolated Sign Language Recognition Dataset
RWTH-PHOENIX-Weather	RWTH-PHOENIX Weather Broadcast German Sign Language Dataset
SIGNUM	SIGNUM German Sign Language Database
SignIT	Italian Sign Language Dataset
SL-ReDu GSL	Sign Language Recognition and Duration Greek Sign Language Corpus
Slovo	Russian Sign Language Dataset
SMILE	Swiss German Sign Language Dataset
SWL-LSE	Spanish Sign Language Word-Level Dataset
WLASL	Word-Level American Sign Language Dataset
YouTube-ASL	YouTube American Sign Language Dataset

1 Introduction

Sign languages are complete natural languages utilized by deaf and hard-of-hearing communities worldwide. Meaning is conveyed through combinations of hand shape, movement, location, and non-manual features such as facial expressions and body posture. Although sign languages serve as the primary mode of communication for millions of individuals globally, they remain substantially underserved by computational tools relative to spoken languages [1, 2].

Computational sign language research aims to develop systems that automatically recognize, translate, and analyze signed communication. Advancement in this field depends on the availability of high-quality datasets, which are crucial for training, evaluating, and benchmarking machine learning models in tasks such as isolated sign language recognition (ISLR), continuous sign language recognition (CSLR), and sign language translation (SLT) [2, 3]. The design characteristics of a dataset, including the sign language represented, recording modality, annotation layers, and evaluation protocol, determine the research tasks it can support and the reliability with which results can be interpreted and compared across studies [1].

Although datasets play a central role in this field, the literature lacks studies that employ a structured and systematic methodology to map and compare dataset characteristics across the research landscape. Previous surveys, such as those by Koller [2], Sarhan and Frintrop [4], and Rastgoo et al. [3], primarily focus on models and methods, treating datasets as supplementary rather than as the primary object of analysis. Ethics-focused studies, such as Bragg et al. [1], address issues of representativeness and fairness in sign language datasets but do not employ a systematic mapping methodology. Structured review studies, such as Wadhawan and Kumar [5], classify recognition systems by method type and modality, yet they do not offer a comprehensive analysis of dataset characteristics. The lack of a structured, dataset-level overview represents a significant gap, as researchers currently lack a consolidated and evidence-based reference for selecting datasets based on task, language, or modality.

This thesis addresses this gap by employing the systematic mapping methodology of Petersen et al. [6, 7], supplemented by the systematic review guidelines of Kitchenham and Charters [8], to produce a structured classification of publicly available sign language datasets. The study identifies, screens, and analyzes 30 datasets using a predefined extraction scheme of 38 attributes, organized into five thematic attribute groups: scale and scope, which addresses language coverage, dataset type, and composition; modality coverage, which captures the data types and recording formats available in each dataset; recording conditions, which documents the environment and technical setup; annotation and benchmarking, which examines labeling depth and evaluation protocols; and signer demographics, which captures participant diversity and reporting completeness. The full data extraction sheets are provided in Appendix A and publicly available via the project repository. The study is guided by the following overarching research question: **How are publicly available sign language datasets designed, and what research directions do their characteristics enable or constrain?** Three sub-questions inform the classification and gap analysis:

1. What structural and technical characteristics define existing sign language datasets, such as language coverage, modality, scale, and annotation type?
2. What types of computational tasks, such as isolated recognition, continuous recognition, translation, and sign-spotting, are supported by these datasets?
3. What patterns, limitations, and research gaps emerge from the current dataset landscape?

The remainder of this thesis is organized as follows. Chapter 2 presents the conceptual background and related work, covering key concepts in sign language computing and situating this study within the existing literature. Chapter 3 describes the systematic mapping methodology and each step of the six-step process. Chapter 4 presents the classification results across all five thematic attribute groups. Chapter 5 discusses the identified gaps, provides recommendations for future dataset development, and offers task-based dataset selection guidelines for researchers. Chapter 6 presents the conclusions of the study.

2 Background and Related Work

In this chapter, we establish a conceptual and scholarly context for the systematic mapping study presented in this thesis. We outline key concepts, prior research, and methodological frameworks that inform the study's design and scope.

2.1 Sign Language Datasets in Computational Research

Sign language datasets serve as the primary empirical foundation for computational sign language research [2, 3]. These datasets comprise video recordings of one or more signers, with each clip annotated to indicate the meaning of the sign or signs being performed. They are utilized to train, evaluate, and benchmark models for a variety of tasks [2].

Existing datasets differ substantially in scope, including the sign language represented, the number of signs and signers, the type of data recorded, and the level of annotation detail [1, 2]. These variations reflect distinct research priorities and directly influence the tasks a dataset can support and the reliability of experimental results [2]. Understanding these characteristics is essential for selecting appropriate datasets and for comparing results across studies [5].

2.2 Types of Sign Language Datasets

A primary distinction among sign language datasets is the type of signing captured: isolated signs, continuous signing, or a combination of both.

An *isolated sign dataset* consists of short video clips in which a single sign is performed independently. Each clip is labeled with the name of the sign, known as a *gloss*, which is a written word used to identify a sign (e.g., WATER, HELLO) as shown in Figure 1. These datasets support sign language recognition (SLR), the task of identifying the sign being performed in a given video clip. A widely used example is WLASL (Word-Level American Sign Language), which contains over 21,000 clips covering 2,000 signs performed by 119 signers [9].



Figure 1: Examples of isolated sign glosses in American Sign Language: HELLO (left) and WATER (right), each performed as a discrete clip and labeled with the corresponding gloss and English translation (Source: https://www.youtube.com/watch?v=stCwx6SXgSA).

A *continuous signing dataset* captures naturally flowing sign language, where multiple signs are produced sequentially to form phrases and sentences, as shown in Figure 2. These datasets are more complex to collect and annotate because sign boundaries must be identified within the video stream. They support tasks such as sign-spotting, which involves detecting where specific signs occur within a continuous signing stream, and sign language translation (SLT), which converts a signed video into a written or spoken sentence in another language. An example is RWTH-PHOENIX-Weather 2014, a German Sign Language dataset of weather forecast broadcasts containing 6,841 signed sentences from 9 signers as illustrated in Figure 2.

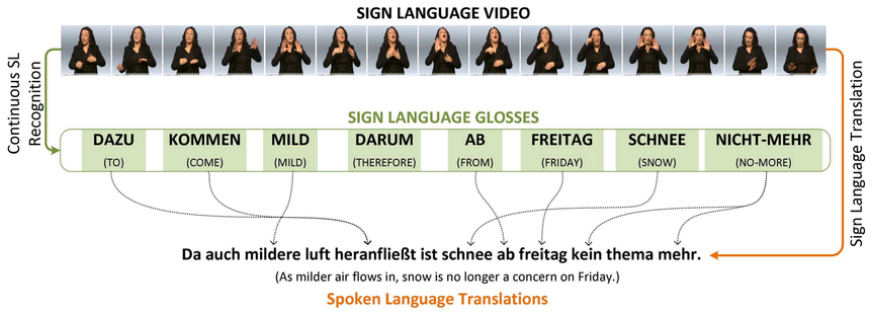


Figure 2: Example of a continuous sign language video from a German Sign Language dataset, illustrating how such datasets support tasks such as sign language translation (SLT). The gloss labels and spoken-language translation shown are in German. (Source: https://www.researchgate.net/figure/Continuous-sign-language-recognition-and-sign-language-translation-2_fig2_389715319).

A third category, referred to as mixed-type datasets, combines isolated signs, continuous sentences, and, in some cases, fingerspelling sequences, as shown in Figure 3, in which individual letters of a written word are represented through distinct handshapes, within a single release. Examples include the SL-ReDu GSL corpus [10], which provides isolated, continuous, and fingerspelling subsets for Greek Sign Language, and AzSLD [11], which covers fingerspelling letters, isolated words, and continuous sentences for Azerbaijani Sign Language. Mixed-type datasets are particularly valuable as they enable models to be evaluated across multiple task types using a consistent signer pool.

2.3 Data Modalities

The term modality refers to the type of sensor or data format used to record signing. Datasets may include one or multiple modalities, each capturing distinct aspects of the signing event.

RGB (Red, Green, Blue) video, which is standard color video from a conventional camera, is the most common modality. It captures the full visual appearance of signing but is sensitive to lighting conditions and background clutter. *Depth data*, which we refer as RGB-D, recorded with sensors such as the Microsoft Kinect, provides

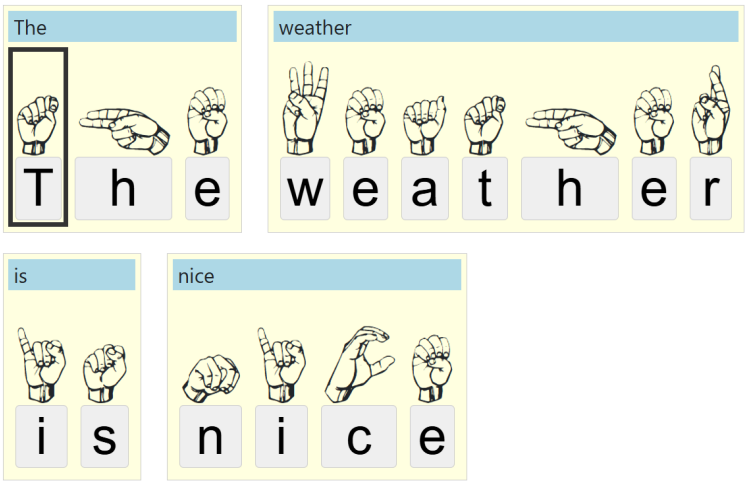


Figure 3: Example of a fingerspelling sequence in which individual handshapes represent the letters of the words "The weather is nice" (Made using the tool https://wecapable.com/tools/text-to-sign-language-converter/).

distance information for each pixel, enabling more robust foreground segmentation and three-dimensional analysis, although it requires specialized hardware. Figure 4 illustrates this contrast using sample frames from How2Sign.



Figure 4: Comparison of RGB video (top row) and RGB-D video (bottom row) recordings from the How2Sign dataset. The faces in these images have been blurred by the dataset authors to protect the privacy of signing participants.(Source: https://how2sign.github.io/).

Skeleton or *pose data* represents the signer as a set of body joint coordinates, such as wrist, elbow, and finger positions, which offers a compact representation that requires significantly fewer model parameters than raw video as shown in Figure 5. These coordinates may be estimated automatically from RGB video using non-intrusive frameworks such as MediaPipe [12] or captured directly through intrusive motion-capture equipment [2]. Pose data is less sensitive to visual factors such as clothing;

however, automatic estimation can be unreliable for rapid hand movements or when one hand obscures the other [13, 14].

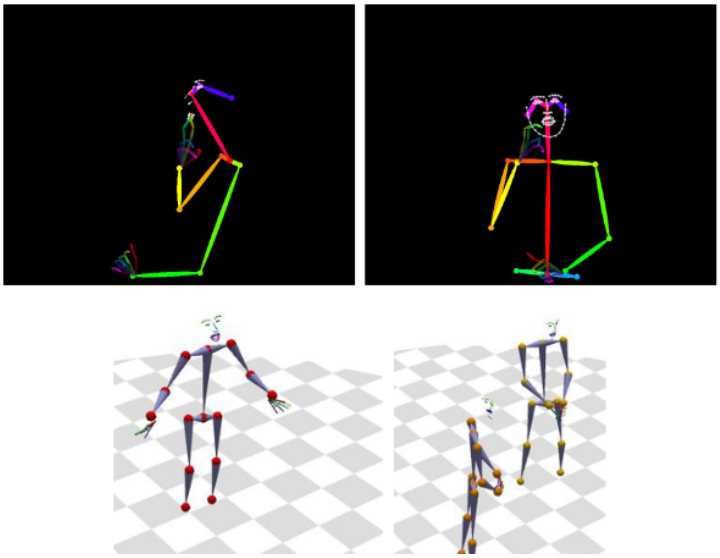


Figure 5: Skeleton or pose representations derived from signing videos in the How2Sign dataset (Source: https://how2sign.github.io/).

Some datasets record signing simultaneously from multiple camera angles, a configuration referred to as multi-view, as shown in Figure 6. In this study, multi-view is treated as a recording configuration rather than a distinct modality, since it refers to the camera arrangement rather than the type of sensor or data format used. Multi-view reduces occlusion but substantially increases the complexity of data collection.

A small number of datasets additionally include audio recordings as a supplementary attribute. Among the datasets included in this study, audio is rarely present and is not used as a primary data source for sign language recognition tasks.

2.4 Annotation in Sign Language Datasets

An *annotation* is a descriptive label assigned to raw video data. The depth of annotation, meaning how many layers of labeling a dataset provides, such as gloss labels, sentence-level translations, or phonological descriptions, determines the range of tasks a dataset can support.

The most common form of annotation is the gloss label, which assigns a sign name to each clip or segment. In continuous datasets, annotations must also specify the start and end points of each sign, a process known as temporal segmentation, which is necessary for sign-spotting tasks. Datasets intended for SLT additionally provide a spoken-language translation of each signed phrase, pairing signed video with a corresponding written sentence (for example, PHOENIX-2014-T includes German sentence translations alongside gloss labels) [15]. A small number of datasets include phonological annotations that describe the sub-lexical structure of signs, such

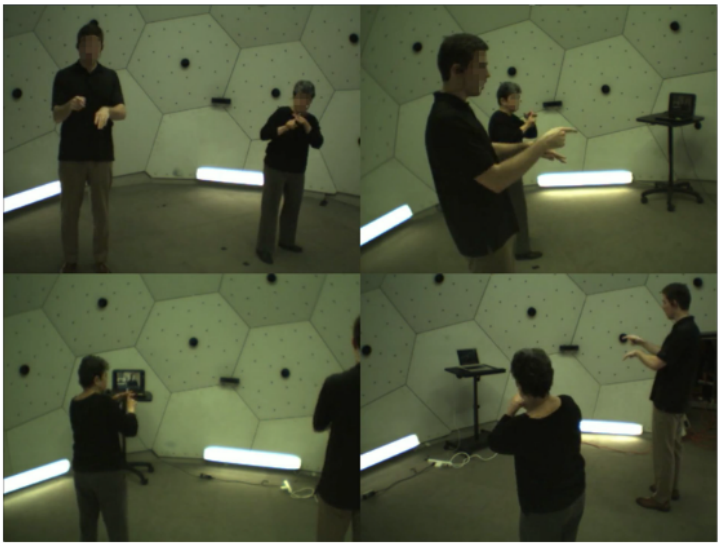


Figure 6: Multi-view recording environment from the How2Sign dataset (Source: https://how2sign.github.io/).

as hand shape, location, and movement, primarily for linguistic analysis rather than recognition.

2.5 Dataset Scale and Signer Diversity

The *scale* of a dataset is typically measured by vocabulary size (the number of distinct sign classes), the total number of video samples, and the number of signers. These dimensions collectively determine the extent of variation within the dataset and the degree to which a trained model can generalize to previously unseen signing.

A larger vocabulary and a greater number of samples per sign generally support more robust model training. The number of signers is particularly important; a dataset with only five signers exposes the model to a limited range of signing styles, which can artificially inflate performance when tested on the same individuals. Datasets with more signers, such as AUTSL (43 signers) [13] or WLASL (119 signers) [9], offer greater diversity, although they remain limited compared to the full breadth of a signing community.

Demographic factors such as age, gender, handedness, and whether signers are deaf native users or hearing learners also influence the representativeness of a dataset. However, this information is often absent or inconsistently reported, which complicates the assessment of potential bias in trained models.

Geographic and linguistic concentration represents a further source of imbalance in the existing dataset landscape. Bragg et al. [1] observe that existing datasets are heavily concentrated in a small number of sign languages, with many of the world's sign languages remaining without dedicated computational resources. Koller [2] similarly notes that large-vocabulary continuous recognition benchmarks have historically been

limited to a small set of languages, most notably German Sign Language, as seen in RWTH-PHOENIX-Weather [16].

2.6 Benchmarking and Evaluation Protocols

Benchmarking involves evaluating a model on a standardized test to facilitate comparison across studies. The evaluation protocol, which defines how the dataset is divided into training and test portions, has a significant impact on reported performance and its practical interpretation.

In a signer-dependent evaluation, some signers appear in both the training and test sets, meaning the model has already been exposed to each test signer's style, which can inflate accuracy. In contrast, a signer-independent evaluation ensures that test signers are absent from the training set, providing a more realistic measure of generalization.

Signer-independent protocols are widely regarded as the more rigorous standard but require datasets with a sufficient number of signers to allow for separation across splits. When a fixed train, validation, and test split is published alongside the dataset as a standardized benchmark, all research groups can evaluate under identical conditions, making results directly comparable. In the absence of such a standard, different groups may partition the same data differently, resulting in outcomes that are difficult to compare. Performance is typically evaluated using standardized metrics. In isolated sign recognition, the most commonly used metric is Top-1 accuracy, which measures how often a sign recognition system correctly identifies the sign performed in a video clip. For continuous signing tasks, Word Error Rate (WER) is employed to measure the proportion of incorrectly predicted words in a signing sequence relative to the reference transcription.

2.7 Dataset Accessibility and Licensing

Accessibility refers to whether a dataset is publicly available, whether it requires a formal application, and the licensing terms that govern its use. Many sign language datasets are distributed under research-only licenses; some are open, while others are restricted to academic institutions. A small number have become unavailable over time due to institutional changes or data protection regulations.

The quality of documentation also affects practical usability. A well-documented dataset includes descriptions of the collection procedure, signer demographics, annotation methodology, file formats, and baseline evaluation results. Inconsistent or incomplete documentation hinders the reproducibility of experiments and limits a dataset's adoption within the research community.

2.8 Related Work

In the following, we explore previous efforts to systematically map and review sign language datasets.

2.8.1 Surveys of Sign Language Recognition and Translation

Several surveys have examined computational sign language research, although their primary focus remains on methods and algorithms rather than datasets. Koller [2] presents a large-scale meta-analysis of sign language recognition, covering approximately 300 published papers and over 400 experimental results from 1983 to 2020. A key finding relevant to this thesis is that a single dataset, RWTH-PHOENIX-Weather [16], served as the only large-vocabulary continuous sign language recognition benchmark worldwide at the time, illustrating the uneven progression of dataset development across task types.

Sarhan and Frintrop [4] review isolated sign language recognition methods over the preceding decade, addressing modality types, feature extraction approaches, and transfer learning strategies. While they provide a detailed overview of the most prominent isolated sign language recognition (ISLR) benchmarks, their scope is intentionally restricted to eight high-profile datasets from the last decade and explicitly refrains from including smaller or non-standard datasets. A subsequent review by Tan et al. [14] expands the scope to include sign language translation and generation, and includes a dedicated analysis of nineteen major datasets. They identify substantial variability in annotation conventions as a primary obstacle to developing robust, generalizable systems.

Rastgoo et al. [3] conduct a deep survey of vision-based deep learning approaches to sign language recognition, covering features, modalities, evaluation metrics, and datasets across five years of published research. Datasets are discussed by modality type and scale and used to benchmark reported model performance. While datasets are categorized by basic annotation targets (e.g., hand or face), deeper characteristics such as annotation granularity, signer-independence verification protocols, and the quality of provided documentation are not analyzed as primary comparative dimensions. Wadhawan and Kumar [5] undertake a decade-long systematic literature review of sign language recognition systems, classifying papers by method type, modality, and application domain. Their study is one of the few in the field to apply a structured review methodology, and it acknowledges variability in dataset properties across studies, but does not treat dataset characteristics as the primary object of structured analysis.

While these surveys collectively establish the importance of dataset metadata such as vocabulary size and signer diversity, their analytical frameworks remain primarily system-centric, which employs datasets as benchmarks for model performance. Consequently, there remains a lack of research that treats sign language datasets as the primary object of structured architectural and qualitative analysis.

2.8.2 Studies Directly Using Sign Language Datasets

Several influential papers focus primarily on introducing or utilizing specific sign language datasets, thereby demonstrating the research community's reliance on well-constructed datasets. Li et al. [9] introduce WLASL, a large-scale American Sign Language dataset containing 2,000 sign classes and recordings from 119 signers, and

show that scale and signer diversity significantly influence recognition performance. Sincan and Keles [13] present AUTSL, a multimodal Turkish Sign Language dataset with both RGB and depth modalities recorded from 43 signers, and provide baseline benchmark results, serving as a concrete example of the benchmarking practices discussed in Section 2.6.

Camgöz et al. [17] utilize the PHOENIX-2014T dataset to develop a transformer-based model capable of performing both continuous sign recognition and sign language translation within a single framework, demonstrating that annotation depth, specifically the availability of both gloss labels and sentence translations, directly enables more advanced research tasks. Duarte et al. [18] introduce How2Sign, a large-scale continuous ASL dataset featuring more than 35,000 signed sentences and a 16,000-word vocabulary across multiple modalities, illustrating the scale and diversity necessary to support real-world translation research.

Bragg et al. [1] adopt a different perspective, examining sign language datasets through the lens of Fairness, Accountability, Transparency, and Ethics (FATE). Their analysis reveals that all existing datasets include only a small number of signers, demographic information is rarely reported, and the rights and responsibilities of data contributors are insufficiently addressed. This work most closely aligns with the present thesis in that it subjects sign language datasets, rather than the models trained on them, to critical scrutiny.

2.8.3 Systematic Mapping Methodology

The methodology underpinning this thesis draws on established guidelines for systematic reviews and systematic mapping studies in software engineering. Kitchenham and Charters [8] provide foundational guidelines for conducting systematic literature reviews, adapted from medical research practices to address the specific challenges of software engineering. These guidelines cover the three phases of a review: planning, conducting, and reporting.

Petersen et al. [6] introduce systematic mapping studies as a distinct approach from systematic literature reviews. While a systematic review aims to answer specific research questions through in-depth analysis of selected studies, a mapping study aims to produce a broad overview of a research area by classifying publications according to predefined attributes and identifying patterns, frequencies, and gaps. Petersen et al. [6] note that systematic mapping is particularly appropriate when a field is large and heterogeneous, precisely the conditions that characterize the sign language dataset landscape. Petersen et al. [7] subsequently provide revised guidelines that update and clarify the original framework.

2.8.4 Positioning This Thesis

The related work reviewed above identifies a persistent gap: although sign language datasets are widely used and their limitations frequently acknowledged, no existing study provides a structured, systematic mapping of dataset characteristics across the field. Surveys treat datasets as supporting material, dataset papers emphasize individual

contributions, and ethics-oriented analyses, such as Bragg et al. [1], while valuable, do not employ a systematic mapping methodology. Structured review studies, such as Wadhawan and Kumar [5], classify methods rather than datasets, leaving dataset characteristics without a dedicated structured treatment. This thesis addresses that gap by applying the guidelines of Petersen et al. [6, 7] and Kitchenham and Charters [8] to produce a structured overview of sign language datasets classified by language coverage, modality, scale, annotation, and accessibility. We detail our methodologies in the next chapter.

3 Methodology

This chapter outlines the methodology used to identify, select, analyze, and classify sign language datasets in this study. The approach follows the systematic mapping method proposed by Petersen et al. [6] and refined in Petersen et al. [7], supplemented by the systematic review guidelines of Kitchenham and Charters [8]. Each section below corresponds to one of the six steps of the systematic mapping process as defined by Petersen et al. [6].

3.1 The Systematic Mapping Method

A systematic mapping study is a form of secondary research that produces a structured overview of a research area by classifying existing publications or artifacts according to predefined attributes. The method was introduced by Petersen et al. [6] and refined in Petersen et al. [7], drawing on the systematic review tradition established in medical research and adapted for software engineering by Kitchenham and Charters [8].

The primary aim of a systematic mapping study is to establish the structure of a research field rather than to synthesize quantitative findings, which is the purpose of a systematic literature review. Systematic mapping identifies which topics have been studied, their frequency, the methods used, and existing research gaps. Petersen et al. [6] describe the output as a visual or tabular map that distributes publications across classification categories derived from the research questions.

3.1.1 Systematic Mapping vs. Systematic Literature Review

Petersen et al. [6] explicitly distinguish between systematic mapping studies and systematic literature reviews, as these approaches are frequently conflated. A systematic literature review is a detailed, focused investigation that aggregates and synthesizes evidence to answer specific research questions, such as "which machine learning algorithm achieves the highest accuracy for isolated sign recognition?" In contrast, a systematic mapping study adopts a broader, classification-oriented perspective, addressing questions such as "what types of datasets exist, and how are they distributed across languages, modalities, and tasks?"

According to Petersen et al. [7], systematic mapping is most suitable when the research field is large and heterogeneous, when the objective is to identify trends and gaps rather than to synthesize quantitative evidence, and when the scope and structure of the field have not yet been comprehensively characterized. These conditions are reflected in this study: the sign language dataset landscape encompasses numerous languages, modalities, and task types; the primary aim is classification and gap identification rather than performance synthesis; besides, the existing surveys reviewed in Section 2.8 address datasets as supporting material rather than as the primary object of structured analysis.

3.1.2 Rationale for Choosing Systematic Mapping

The selection of a systematic mapping methodology, rather than a narrative or systematic literature review, was determined by the specific characteristics of the research area and the objectives of the study.

A narrative review is unsuitable in this context because it lacks a structured search process and predefined selection criteria, which increases susceptibility to selection bias and limits replicability. A systematic literature review would require a narrower research question and would aim to synthesize performance results across datasets. However, this approach is currently infeasible due to significant differences in evaluation protocols, metrics, and sign languages across studies, which prevent meaningful quantitative synthesis. As observed by Kitchenham and Charters [8], it is often impossible to perform meta-analysis in software engineering research because reporting protocols vary so significantly between studies.

Systematic mapping is the most appropriate approach due to the field's size, heterogeneity, and the lack of prior structured overviews. Petersen et al. [7] note that mapping is especially suitable when understanding the scope and structure of existing work is necessary before conducting more focused investigations. This context applies to sign language dataset research, where the absence of a structured overview has been identified as an impediment to progress.

3.1.3 The Six-Step Process

Petersen et al. [6] define systematic mapping as a six-step process as shown in Figure 7. Each step is briefly described below.

Step 1 - *Define research questions.* The process begins with the formulation of research questions that determine the scope of the mapping and guide the classification scheme. In contrast to the typically narrow and empirical questions of SLRs, mapping research questions are broad and classificatory, such as "what types of X exist?" or "how is the field distributed across Y?" Petersen et al. [6] recommend that research questions correspond directly to the classification categories used in the subsequent data extraction step.

Step 2 - *Conduct the systematic search.* A keyword-based search is performed across multiple academic databases to ensure comprehensive coverage. Petersen et al. [6] recommend combining several search terms with Boolean operators and searching at least two to three major databases. The search process should be documented in sufficient detail to enable replication. Backward snowballing, which involves examining reference lists of included papers to identify additional relevant sources, is also recommended as a complementary strategy.

Step 3 - *Screen publications.* All publications retrieved through the search are screened using predefined inclusion and exclusion criteria. Petersen et al. [6] specify a two-stage screening process: an initial screen based on title and abstract, followed by a full-text review of the remaining records. The same criteria must be applied at both stages, and these criteria should be defined prior to screening to prevent post-hoc adjustments that could introduce bias. Kitchenham and Charters [8] further emphasize

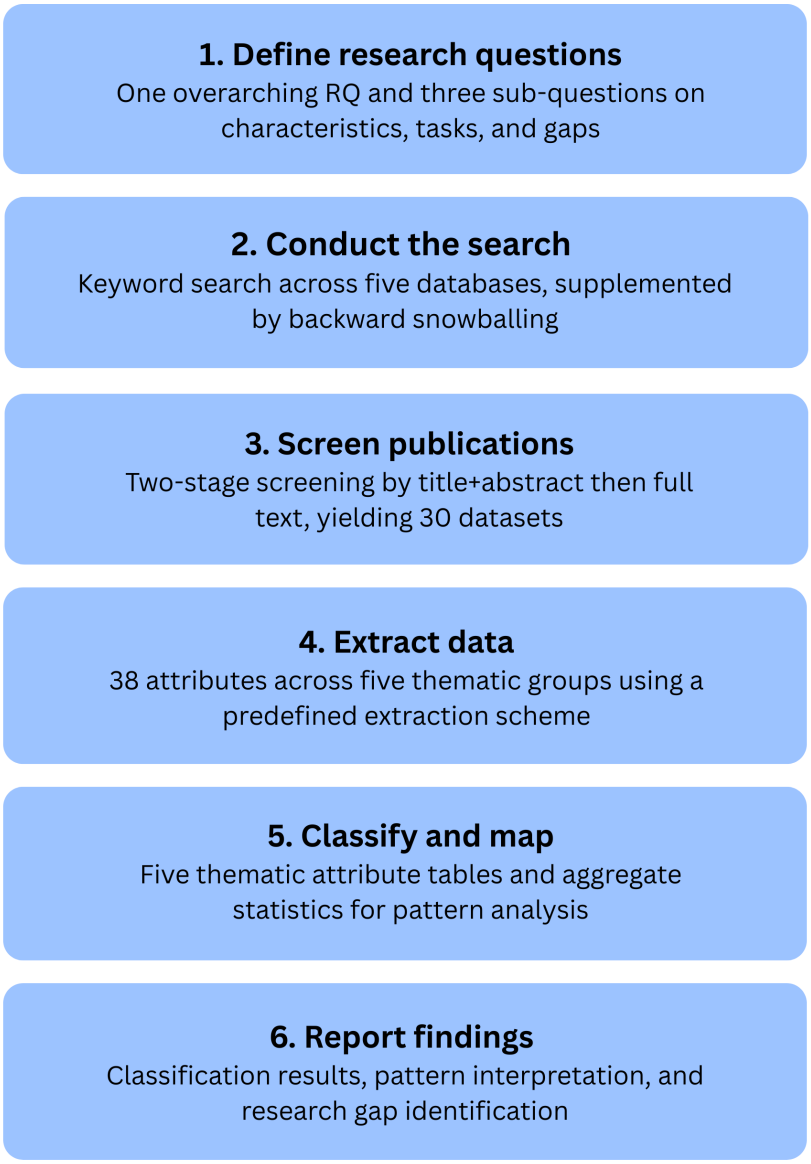


Figure 7: The six-step systematic mapping process as applied in this study, adapted from Petersen et al. [6].

the importance of transparently reporting the number of records retained and excluded at each stage.

Step 4 - *Extract data using a predefined keywording scheme.* In contrast to systematic reviews, which extract quantitative results, systematic mapping studies focus on extracting classification-relevant metadata from each included publication. Petersen et al. [6] refer to this as "keywording," which involves identifying the attributes of each study that are relevant to the research questions. A structured extraction form or spreadsheet is used to ensure consistent capture of attributes across all included studies.

Step 5 - *Classify and map the results.* The extracted data are used to classify publications across the dimensions identified in Step 1. Petersen et al. [6] present classification results as frequency counts and visual maps that illustrate the distribution of studies across categories, revealing the structure of the research area and facilitating the identification of patterns and concentrations.

Step 6 - *Report findings.* The final step is to report the classification results, interpret the patterns identified in the map, and explicitly identify research gaps. Petersen et al. [7] emphasize the importance of discussing both existing coverage and gaps, as gap identification is a primary contribution of a mapping study.

The six-step systematic mapping process was applied in this study, from the initial formulation of research questions through to the reporting of classification results and research gap identification, as shown in Figure 7. Each step is described in detail in Sections 3.2 through 3.7.

3.2 Step 1: Research Questions

Following Step 1 of the Petersen et al. [6] process, the research questions were formulated to be broad and classificatory, aligned with the mapping objective. The following overarching question and sub-questions were established prior to the commencement of the search:

Research Question: *How are publicly available sign language datasets designed, and what research directions do their characteristics enable or constrain?*

Sub-question 1: *What structural and technical characteristics define existing sign language datasets (such as language coverage, modality, scale, and annotation type)?*

Sub-question 2: *What types of computational tasks (such as isolated recognition, continuous recognition, translation, and sign spotting) are supported by these datasets?*

Sub-question 3: *What patterns, limitations, and research gaps emerge from the current dataset landscape?*

These questions align with the thematic attribute tables described in Section 3.6.1 and the structure of the results in Chapter 4. Sub-questions 1 and 2 inform the classification categories, while Sub-question 3 informs the gap analysis.

3.3 Step 2: Search Strategy

This section outlines the data sources and search methodology employed to identify sign language datasets for inclusion in the study. The search strategy integrated keyword-based database queries with backward snowballing to ensure comprehensive coverage of both established and recently published datasets.

3.3.1 Data Sources

The search was conducted across five primary academic databases: Google Scholar, IEEE Xplore, ACM Digital Library, SpringerLink, and ScienceDirect. These databases were selected for their comprehensive coverage of computer science, computer vision, and natural language processing research, which are the main disciplines in which

sign language datasets are developed and published. Project websites, institutional dataset repositories, and research group pages were also manually searched to identify datasets documented beyond traditional academic venues.

3.3.2 Search Procedure

A keyword-based search strategy was implemented using an iterative, progressively narrowing approach. The search started with broad umbrella terms to capture a wide perspective and identify the most recognized and frequently cited datasets. The scope was then incrementally narrowed along four dimensions: geographic region, individual country, dataset type, and computational task.

For geographic narrowing, searches moved from continental terms to country-specific terms and then to country-specific acronyms. For example, beginning with "European sign language dataset," the search would narrow to "Finnish sign language dataset" and then to "FinSL dataset" and "FinSL corpus." This approach was applied to all regions and countries represented in the search. For the dataset type, the terms "isolated sign language dataset" and "continuous sign language dataset" were used. For task type, the search used terms such as "sign language recognition dataset," "sign language translation dataset," "sign language recognition corpus," and "sign language translation corpus." Throughout all iterations, both long-form and abbreviated variants were used, such as "sign language dataset" and "SL dataset," or "sign language corpus" and "SL corpus."

The search terms were combined using Boolean operators to construct queries suited to each database's indexing requirements. The general structure applied was: ("sign language" OR "SL") AND ("dataset" OR "corpus") AND ("isolated" OR "continuous" OR "recognition" OR "translation"). This structure accommodates both long-form and abbreviated variants used throughout the search and captures the two primary dataset types and two primary computational tasks targeted by this study.

For each search query, results were reviewed up to the second page. Results beyond page two were examined only when the title or URL contained terms such as "dataset," "corpus," "recognition," or "translation."

The search was supplemented by backward snowballing, which involved reviewing the related work sections and reference tables of all included papers to identify additional relevant datasets. This method is consistent with the guidelines of Petersen et al. [7] for identifying additional relevant sources. Snowballing concluded when five consecutive papers produced no new eligible datasets, a saturation criterion adopted to ensure the search was thorough and unbiased, in accordance with the principles for a well-defined methodology established by Kitchenham and Charters [8].

3.4 Step 3: Screening

The following section outlines the procedures for screening records retrieved during the search, along with the inclusion and exclusion criteria applied at each stage. Records were assessed using predetermined criteria in two sequential stages: initial title and abstract screening, followed by full-text evaluation.

3.4.1 Screening Procedure

The screening process followed the two-stage procedure specified by Petersen et al. [6] and Kitchenham and Charters [8]. In the first stage, titles and abstracts of all retrieved records were reviewed for keywords such as "dataset," "corpus," "recognition," and "translation." Records that clearly described gesture vocabularies, which are predefined sets of hand gestures for human-computer interaction rather than natural sign languages, as well as computer-generated or animated signing data, or records unrelated to natural sign language, were excluded at this stage.

In the second stage, the full texts of the remaining records were evaluated against all inclusion and exclusion criteria specified in Sections 3.4.2 and 3.4.3. Typically, each search query, consisting of a progressively narrowed combination of keywords and Boolean operators applied across the selected databases, produced 10 to 15 candidate records at the title and abstract stage, with 2 to 4 retained after full-text screening. This ratio demonstrates the breadth of the initial search and the specificity of the inclusion criteria. Records that lacked sufficient documentation for attribute extraction or failed to meet the inclusion criteria upon full review were excluded. Figure 8 presents the overall selection process from initial database retrieval through full-text screening and backward snowballing to final inclusion.

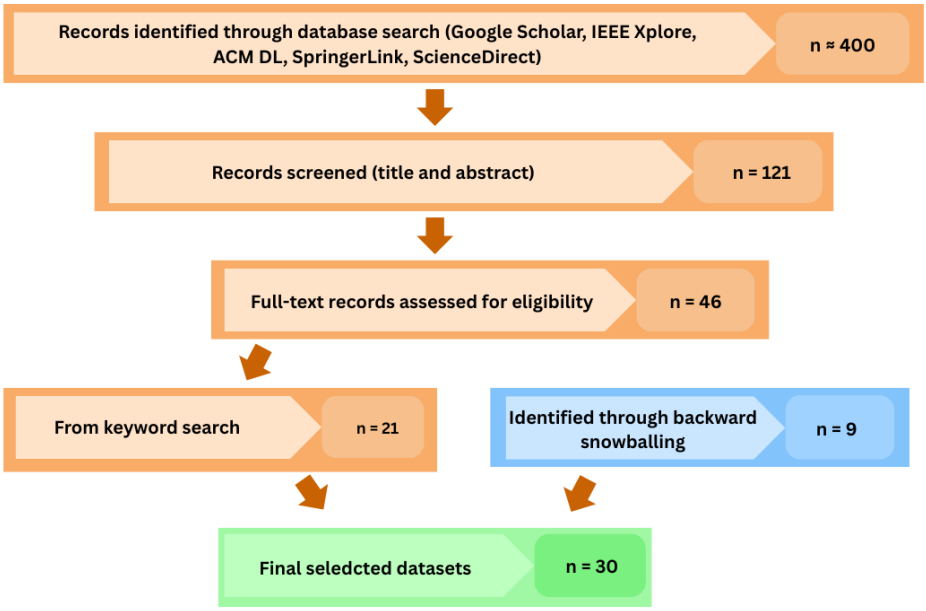


Figure 8: Dataset selection and screening process.

3.4.2 Inclusion Criteria

A dataset was included if it:

1. described a dataset related to a natural sign language, that is, a language used by a deaf or hard-of-hearing community rather than a gesture vocabulary designed

for human-computer interaction;

2. provided visual data in the form of video or image recordings;
3. was documented in a peer-reviewed paper, conference proceeding, or official dataset description; and
4. was documented in English.

3.4.3 Exclusion Criteria

A dataset was excluded if it:

1. focused solely on gesture recognition unrelated to any natural sign language;
2. lacked sufficient documentation to extract the required attributes;
3. consisted solely of synthetic or computer-generated signing data; or
4. was not publicly accessible or identifiable through public documentation.

3.4.4 Selected Datasets

These criteria were defined in the protocol before screening began and applied uniformly across both screening stages, in accordance with the requirements of Petersen et al. [6] and Kitchenham and Charters [8]. This process yielded a final set of 30 datasets for inclusion in the study.

Final selected datasets are: How2Sign [18], MS-ASL [19], WLASL [9], YouTube-ASL [20], OpenASL [21], BOBSL [22], RWTH-PHOENIX-Weather 2014 [23], SIGNUM Database [24], AUTSL [13], BosphorusSign22k [25], Slovo [26], Corpus FinSL [27], RoCoISLR [28], BdSLW60 [29], INCLUDE [30], CE-CNSL [31], CSL-Daily [32], SL-ReDu GSL corpus [10], SMILE [33], MM-WLAuslan [34], LSA64 [35], AzSLD [11], KSU-SSL [36], Alabib-65 [37], Mediapi-RGB [38], SWL-LSE [39], SignIT [40], CASL-W60 [41], KSL-Guide [42], LIBRAS-UFOP [43].

The full list of included datasets, together with their key attributes, is presented in the Data Extraction Sheet A.1 in Appendix A.

3.5 Step 4: Data Extraction

This section describes the attribute selection process and the extraction procedure used to collect data from each of the 30 included datasets. The extraction scheme was established prior to data collection to ensure consistency and to prevent post-hoc adjustments.

3.5.1 Attribute Selection

Consistent with Step 4 of the Petersen et al. [6] process, a predefined extraction scheme was established before data collection. An initial list of approximately 50 to 60 candidate attributes was compiled based on the research questions and the thematic attribute tables described in Section 3.6.1, alongside the anticipated needs of researchers working with sign language datasets. This list was iteratively refined by evaluating each candidate attribute against the research sub-questions and the thematic groupings established for this study, and removing attributes that did not directly address any sub-question, until a final set of 38 attributes was reached: 35 core descriptive attributes and 3 supplementary fields for unique features, limitations, and notes.

The 35 core attributes are: Dataset Name, Language(s), Country/Region, Year Published, Authors, Institutions/Affiliations, Publicly Available, License Type, Citation, Dataset Download Link, Number of Signs/Classes, Number of Videos/Samples, Total Hours, Dataset Type, RGB Video, Depth Video, Skeleton/Pose Data, Multi-view Cameras, Audio Included, Recording Environment, Background Complexity, Lighting Conditions, Camera Type, Resolution, Frame Rate, Signer-Independent Split, Baseline Model Results, Gloss Labels, Sentence Alignment, Non-Manual Annotation, Number of Signers, Deaf/Hearing/CODA Representation, Gender Distribution, Age Range, and Dialect/Regional Variation.

3.5.2 Extraction Process

Data extraction proceeded in two stages. First, attribute values were recorded in a 'Data Extraction Sheet' (Appendix A.1) exactly as stated in the source documentation, without interpretation. Second, the extracted data were standardized and cleaned to ensure uniform formatting across all 30 datasets (Appendix A.2). Standardization included removing extraneous white space, normalizing variant expressions of the same value into a single canonical form (such as consolidating "Not explicitly stated," "Not mentioned," and "Not explicitly listed" to "Not specified"), correcting formatting inconsistencies, and ensuring consistent use of units and parentheses. When documentation was ambiguous or missing, the value was recorded as 'Not specified' to maintain accuracy and avoid bias, in line with Kitchenham and Charters [8]. In the thematic attribute tables presented in Appendix B, 'Not specified' is represented by a dash (—) for typographic economy; the two forms are equivalent in meaning throughout the study.

3.6 Step 5: Classification and Mapping

This section outlines the organization, classification, and analysis of the extracted data. Classification results are provided in two complementary formats: five thematic attribute tables addressing the dimensions most pertinent to the research questions, and an aggregate analysis of patterns and coverage rates derived from these tables. Sections 3.6.1 and 3.6.2 provide detailed explanations of each approach.

3.6.1 Thematic Attribute Tables

In accordance with Step 5 of the Petersen et al. [6] process, the extracted attributes were categorized into seven thematic attribute tables: General Information, Scale and Scope, Modalities, Recording Conditions, Annotation and Benchmarking, Signer Demographics, and Remarks. Table 1 presents the correspondence between each thematic attribute table and the respective research sub-questions.

Table 1: Mapping of thematic attribute tables to research sub-questions

Thematic Attribute Table	SQ1: Structural & technical characteristics	SQ2: Supported computational tasks	SQ3: Patterns, limitations & gaps
Scale and Scope	✓		✓
Modalities	✓		
Recording Conditions	✓		
Annotation and Benchmarking	✓	✓	✓
Signer Demographics	✓		✓

Consistent with the tabular map output described by Petersen et al. [6], classification results are presented in Chapter 4 in two complementary forms: five thematic attribute tables (Tables B1, B2, B3, B4, B5) that address the dimensions most relevant to the research questions, and an aggregate analysis of patterns and coverage rates derived from these tables. The five thematic attribute tables are presented as follows:

Scale and Scope: Table B1 presents, for all 30 datasets, the dataset name, year, type, number of sign classes, number of video samples, total duration in hours, and number of signers.

Modality Coverage: Table B2 displays the availability of RGB video, depth video, skeleton/pose data, multi-view cameras, and audio for each dataset.

Recording Conditions: Table B3 presents the recording environment, background complexity, lighting conditions, resolution, and frame rate for each dataset.

Annotation and Benchmarking Characteristics: Table B4 includes information on gloss labels, sentence alignment, non-manual annotation, signer-independent split, and baseline results.

Signer Demographics: Table B5 presents the number of signers, deaf, hearing, or Children of Deaf Adults (CODA) status, gender distribution, age range, and dialect or regional variation. A summary of completeness at the bottom indicates how many datasets report each demographic field.

The remaining two thematic attribute groups, General Information and Remarks, are not represented as standalone thematic attribute tables, but are present in the Data Extraction Sheets A. Key fields from General Information, such as dataset name, language, country, year, and availability, are incorporated into Table B1. The Remarks group contains supplementary fields for dataset-specific unique features, limitations, and notes that are not suitable for structured comparison.

3.6.2 Attribute Analysis

The classification results recorded in the thematic attribute tables were used to conduct a structured analysis addressing the research sub-questions. For each thematic attribute table, aggregate statistics, frequency counts, and coverage rates were calculated to characterize the dataset landscape and identify patterns. For Scale and Scope (Table 4), the median, minimum, and maximum values were calculated for duration, signer count, vocabulary size, and video sample count, with the number of reporting datasets noted for each metric. Aggregate counts, including the total number of datasets, the number of unique sign languages represented, and the combined documented duration, were also derived. For Modality Coverage (Table 5), the counts of Yes, No, and Partial responses were summed for each modality across all 30 datasets, and percentage coverage rates were computed. For Annotation and Benchmarking (Table 6), an annotation score was calculated for each dataset by assigning 1 point for each "Yes" response, 0.5 points for each "Partial" response, and 0 points for each "No" response across the five annotation features, resulting in a maximum possible score of 5.0. For Signer Demographics (Table 7), a completeness summary was derived by counting the number of datasets that reported each demographic field. For Geographic and Linguistic Coverage (Table 3), datasets were grouped by region to determine the distribution of represented sign languages. These aggregate analyses, combined with the dataset-level details in each thematic attribute table, form the basis of the results presented in Chapter 4 and the gap analysis in Chapter 5.

3.7 Step 6: Reporting of Classification Results

In accordance with Step 6 of the Petersen et al. [6] process, classification results are presented in Chapter 4 and interpreted in Chapter 5. Chapter 4 presents the structured findings of the thematic attribute tables outlined in Section 3.6.1, supported by the aggregate analysis described in Section 3.6.2. Frequency counts, coverage rates, and distributional patterns are employed to characterize the current state of the sign language dataset landscape.

Chapter 5 undertakes the interpretive analysis required by this step, addressing Sub-question 3 by identifying patterns, limitations, and research gaps that emerge from the classification results. Consistent with Petersen et al. [7], which emphasizes gap identification as a primary contribution of mapping studies, the discussion clearly distinguishes between areas of relative coverage and those where the dataset landscape is sparse or structurally limited. The findings in both chapters are based exclusively on extracted attribute data and source documentation, in accordance with the evidence-based requirements of the systematic mapping methodology.

3.8 Validity Considerations

Petersen et al. [6] identify several threats to validity that are common in systematic mapping studies and Kitchenham and Charters [8] provide corresponding mitigation strategies. The risk and mitigation that apply to this study are shown in Table 2.

Table 2: Risk to validity and mitigation strategies

Risk to Validity	Mitigation Strategy
Search incompleteness: relevant datasets may be absent from the selected databases, documented only in non-English sources, or distributed through informal channels	This risk was mitigated by searching five academic databases in parallel, supplementing the keyword search with manual searches of project websites and institutional repositories, and applying backward snowballing until saturation was reached.
Selection bias: subjective judgments during title and abstract screening may introduce inconsistency	The inclusion and exclusion criteria were defined in the protocol before screening began and applied uniformly across both screening stages.
Extraction inconsistency: ambiguous or incomplete dataset documentation may lead to inconsistent attribute recording	All ambiguous values were recorded as "Not specified" rather than estimated, and the same extraction scheme was applied to every dataset.
Classification subjectivity: the assignment of datasets to classification categories involves judgment	Classification dimensions were defined before extraction began and are directly derived from the research questions, reducing the scope for post-hoc adjustment.

These mitigation strategies are consistent with the recommendations of Petersen et al. [6, 7] and Kitchenham and Charters [8], and are documented to enhance the reproducibility and transparency of this study.

3.9 Tools and Platforms

Several software tools and online platforms were used throughout this study to support different stages of the work, from data collection to manuscript preparation. Dataset identification utilized five academic databases: Google Scholar, IEEE Xplore, ACM Digital Library, SpringerLink, and ScienceDirect. Data extraction and verification were supported by Google NotebookLM, which enabled the review of dataset documentation and identification of relevant attribute values from original sources. Extracted data were recorded, organized, and analyzed in Google Sheets, which served as the primary environment for the classification spreadsheet. Grammarly was employed to check spelling, grammar, and sentence structure throughout the writing process. Bibliographic records for all cited sources were managed using Zotero, which also exported references as BibTeX files for integration with the LaTeX manuscript. The initial manuscript draft was prepared in Google Docs, while the final thesis was formatted in Overleaf using LaTeX for document formatting, citation management, and consistent presentation. Claude supported brainstorming, idea organization, concept

clarification, structural planning, and LaTeX formatting. Figures and diagrams were created and edited using Canva. The geographic distribution map in Chapter 4 was produced using MapChart, which highlights countries on a world map. Version control for data extraction sheets was maintained using Git, with files hosted on GitHub.

3.10 Summary

Taken together, the six steps described in this chapter establish a transparent and replicable procedure for the identification, selection, extraction, and classification of the 30 datasets included in the study. The methodological choices made at each step are grounded in the guidelines of Petersen et al. [6, 7] and Kitchenham and Charters [8], and are documented to support both the credibility of the findings and their reproducibility by future researchers. Chapter 4 presents the classification results produced by applying this procedure.

4 Results

The results of the systematic mapping study described in Chapter 3 are presented in this chapter. Thirty publicly available sign language datasets were identified, selected, and analyzed using a predefined 38-attribute extraction scheme. The findings are presented across the five thematic attribute groups established in Section 3.6.1: scale and scope, modality coverage, recording conditions, annotation and benchmarking, and signer demographics. Thematic attribute tables in Appendix B (B1, B2, B3, B4, and B5) provide full dataset-level detail for each group, while Tables 3, 4, 5, 6, and 7 present aggregate counts and summary statistics to support interpretation.

4.1 Overview of Included Datasets

Thirty sign language datasets were identified and included through the systematic search, screening, and selection process described in Chapter 3. These datasets represent 23 distinct sign languages, originate from five geographic regions, and were published between 2010 and 2025. Collectively, they provide a documented duration of approximately 3,697 hours across the 19 datasets for which this information is available. All datasets are publicly available or accessible for research purposes. Table B1 lists the included datasets, along with their publication year, type, number of sign classes, number of video samples, total duration, and number of signers.

4.2 Distribution by Sign Language and Region

The 30 datasets encompass 23 distinct sign languages distributed across five geographic regions. Table 3 summarizes the regional distribution.

Table 3: Regional distribution of the 30 included datasets

Region	#Datasets	Sign Languages Covered
Europe	12	British SL, German SL (×2), Finnish SL, Greek SL, Italian SL, French SL, Spanish SL, Russian SL, Romanian SL, Swiss German SL, Azerbaijani SL
Asia-Pacific	9	Turkish SL (×2), Indian SL, Chinese SL (×2), Australian SL, Bangla SL, Saudi SL, Korean SL
North America	5	American SL (×5)
Latin America	2	Argentinian SL, Brazilian SL
Africa	2	Algerian SL, Central African SL

Among the thirty included datasets, twelve originate from Europe and cover eleven distinct sign languages. German Sign Language (DGS) is represented by two datasets: RWTH-PHOENIX-Weather 2014 [23] and SIGNUM [24]. Nine datasets originate from the Asia-Pacific region, spanning seven sign languages, with Turkish

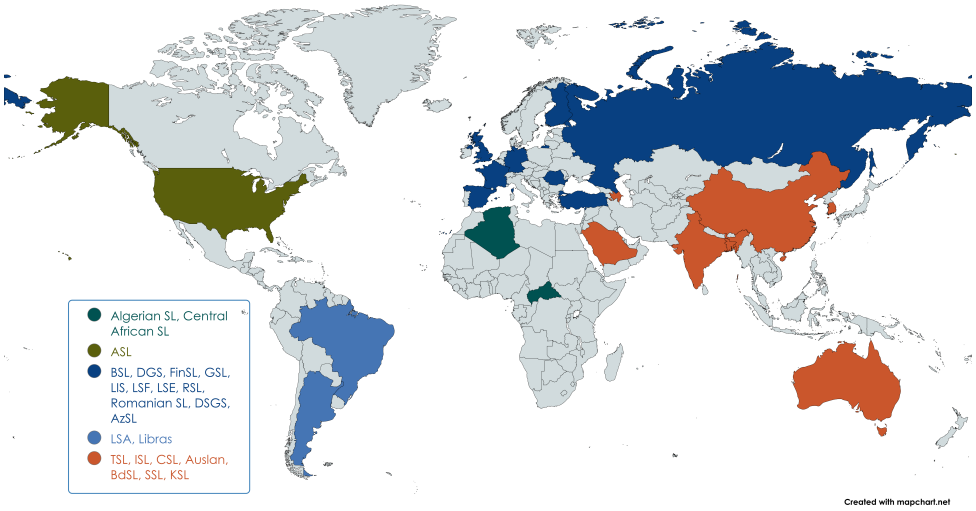


Figure 9: Regional distribution of the 30 included datasets.

Sign Language covered by two datasets (AUTSL [13] and BosphorusSign22k [25]) and Chinese Sign Language by two datasets (CE-CNSL [31] and CSL-Daily [32]). Five datasets originate from North America, all of which target American Sign Language (ASL), making ASL the most represented language in the corpus. Latin America and Africa each contribute two datasets. Notably, datasets for several sign languages, including Indonesian, Japanese, Nigerian, and Pakistani Sign Language, were not identified through the systematic search of this study, which indicate that these languages remain without well-documented publicly available computational resources.

4.3 Dataset Type and Task Support

The 30 datasets are categorized into three types (shown in Table B1): 16 isolated-sign datasets (53%), 9 continuous-sign datasets (30%), and 5 mixed-type datasets (17%). Figure 10 illustrates this distribution.

Isolated sign datasets primarily support sign language recognition (SLR) at the word level. Each dataset consists of short, independently produced sign clips labeled with gloss identifiers. Continuous datasets support more complex tasks, including continuous sign language recognition (CSLR), sign-spotting (locating where specific signs occur within an unsegmented signing stream), and sign language translation (SLT), which require paired spoken-language annotations not present in isolated datasets. Mixed-type datasets offer structured coverage across multiple task types within a single release, which makes them valuable for evaluating models across task boundaries with a consistent signer pool. The predominance of isolated datasets suggests that current research is more strongly supported for word-level recognition tasks than for sentence-level or translation tasks.

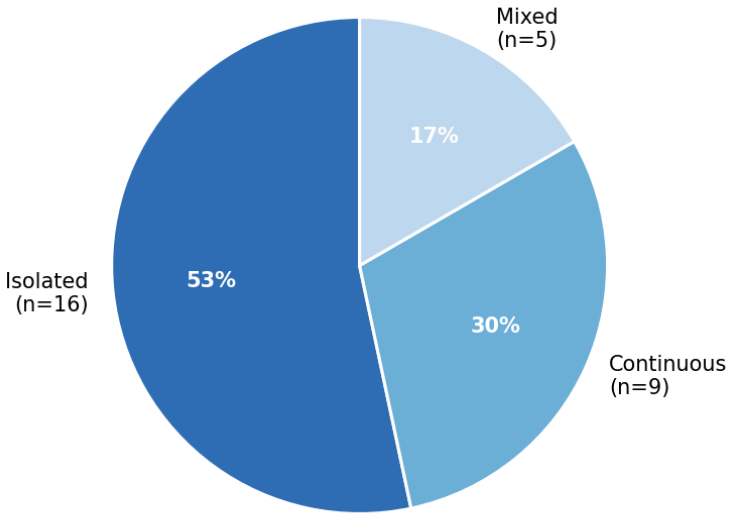


Figure 10: Distribution of the 30 included datasets by type.

4.4 Dataset Scale and Composition

The 30 datasets exhibit substantial variation in scale across four dimensions: duration, number of signers, vocabulary size (noting that the maximum of 107,000 reflects Corpus FinSL's token count rather than a discrete sign-class vocabulary), and number of video samples. Table 4 summarizes the key scale statistics derived from Table B1.

Table 4: Summary scale statistics across the 30 included datasets

Metric	Median	Min	Max	Datasets Reporting
Duration (hours)	24	3	1,467	19 of 30
Number of signers	27	5	2,519	30 of 30
Sign classes/vocabulary	872	56	107,000	30 of 30
Video samples	9,224	343	521,000	30 of 30

At the upper end, BOBSL [22] contains approximately 1,467 hours of footage from 39 signers, making it the largest dataset by duration. YouTube-ASL [20] provides 984 hours of data from more than 2,519 signers. KSL-Guide [42] is the largest dataset by video count, with over 521,000 samples from 72 signers, followed by MM-WLAuslan [34] with over 282,900 samples from 73 signers. At the lower end, SignIT [40] contains 3.33 hours across 644 videos from 45 signers, and INCLUDE [30] provides approximately 3 hours with 4,287 videos from 7 signers. The median signer count of 27 is influenced by a small number of large-scale outliers: YouTube-ASL [20] reports over 2,519 signers, and MS-ASL [19] contributes 222, both considerably higher than the rest of the corpus. Excluding these two datasets reduces the median of the remaining 28 to approximately 25 signers, which more accurately reflects the typical

dataset scale. Most datasets rely on relatively small, closed participant groups, a pattern that has direct implications for model generalization, as discussed in Section 5.1.6. Total duration is not specified for 11 datasets: AUTSL [13], RoCoISLR [28], BdSLW60 [29], SMILE [33], MM-WLAuslan [34], LSA64 [35], Alabib-65 [37], SWL-LSE [39], CASL-W60 [41], KSL-Guide [42], and LIBRAS-UFOP [43].

4.5 Modalities and Recording Setup

The modality coverage across the 30 datasets is summarised in Table 5 and Figure 11. Table B2 presents the full details of the modalities and recording setup.

Table 5: Aggregate modality coverage counts across the 30 included datasets

Modality	Yes	Partial	No	% Yes or Partial
RGB Video	29	1	0	100%
Skeleton / Pose	17	0	13	57%
Depth Video	7	0	23	23%
Multi-view Cameras	5	1	24	20%
Audio	2	1	27	10%

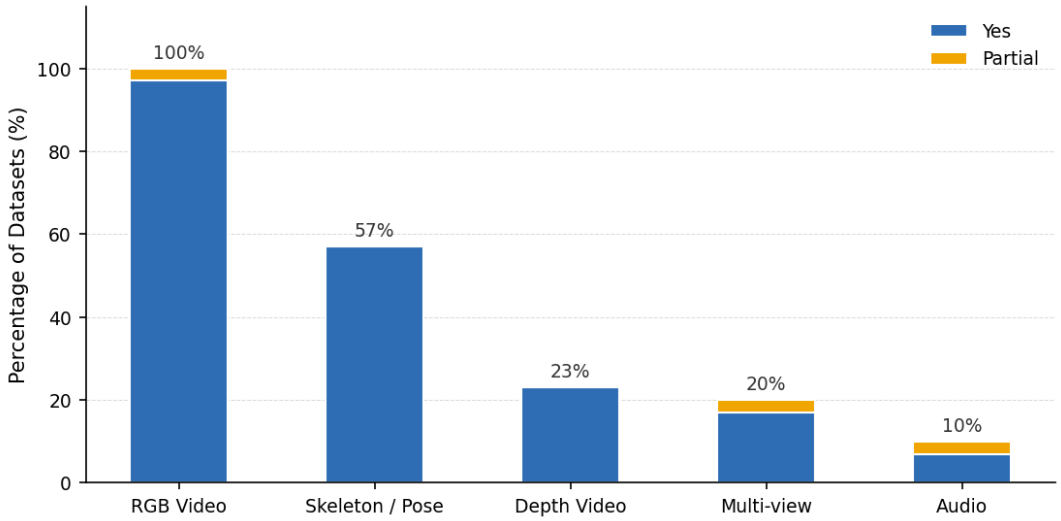


Figure 11: Modality coverage rates across the 30 included datasets.

RGB video is present in 29 of the 30 datasets, with SWL-LSE [39] providing partial coverage. Skeleton or pose data is available in 17 datasets. Depth video is available in 7 datasets: AUTSL [13], BosphorusSign22k [25], SL-ReDu GSL [10], SMILE [33], MM-WLAuslan [34], LIBRAS-UFOP [43], and How2Sign [18]. Multi-view camera setups are present in 5 datasets: MM-WLAuslan [34], How2Sign [18], SMILE [33], AzSLD [11], and KSL-Guide [42]. Audio is available in only How2Sign [18] and RWTH-PHOENIX-Weather 2014 [23], with Mediapi-RGB [38] reporting partial audio. The complete modality details for all 30 datasets are presented in Table B2.

Recording environment details, including background complexity, lighting conditions, camera type, resolution, and frame rate, are presented in full in Table B3 (Recording Conditions).

Multi-view is considered a recording configuration rather than a sensor modality, as it refers to the camera arrangement rather than the type of data captured. However, it is included in Table B2 and Table 5 alongside modality attributes to provide consolidated information for researchers selecting datasets for multi-camera experiments.

4.6 Annotation and Benchmarking Practices

Annotation and benchmarking coverage across the five features examined is summarised in Table 6 and Figure 12. Note that for Signer-Independent Split, two datasets (SL-ReDu GSL [10] and SignIT [40]) do not report this information in their source publications; the percentage is computed across all 30 datasets. Table B4 presents annotation and benchmarking characteristics for all 30 datasets.

Table 6: Aggregate annotation and benchmarking feature coverage across the 30 included datasets, sorted by coverage rate

Annotation Feature	Yes	Partial	No	% Yes or Partial
Gloss Labels	27	0	3	90%
Baseline Results	27	0	3	90%
Signer-Independent Split	16	0	12	53%
Sentence Alignment	15	0	15	50%
Non-Manual Annotation	3	6	21	30%

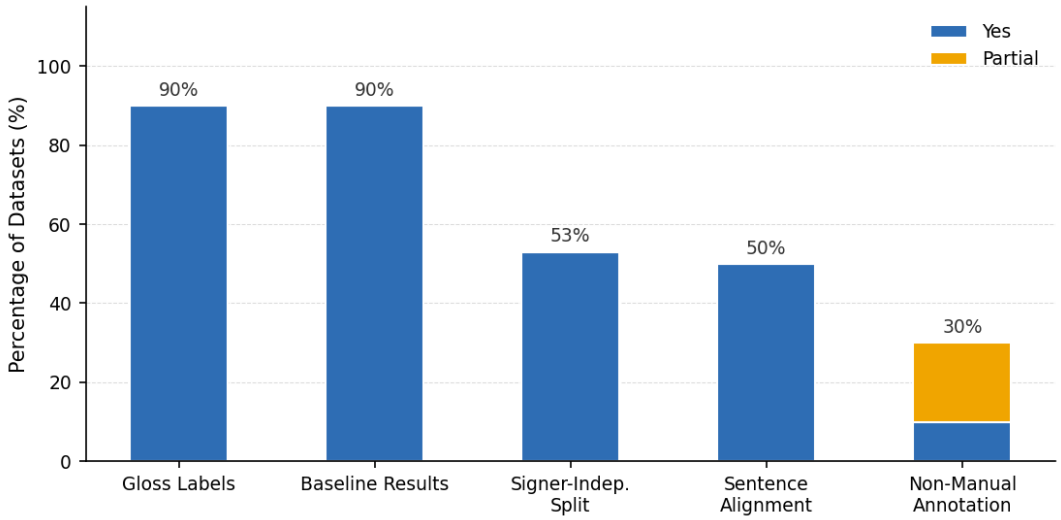


Figure 12: Annotation and benchmarking feature coverage rates across the 30 included datasets.

Gloss labels and baseline results are the most consistently provided features, each present in 27 datasets. Gloss labels are absent from YouTube-ASL [20], OpenASL [21], and Mediapi-RGB [38], which instead provide sentence-level translations as their primary annotation. Baseline results are not specified for Corpus FinSL [27], SMILE [33], and SL-ReDu GSL [10]. Signer-independent splits are defined by 16 datasets, while 12 datasets do not define this protocol, and 2 datasets (SL-ReDu GSL [10] and SignIT [40]) do not report this information. Sentence alignment is present in only 15 of the 30 datasets. Non-manual annotation is the least represented feature: only How2Sign [18], LIBRAS-UFOP [43], and SL-ReDu GSL [10] fully annotate non-manual features, while 21 datasets provide none. To support comparison across datasets, an annotation score is computed for each dataset by assigning 1 point for each 'Yes' response, 0.5 points for each 'Partial' response, and 0 points for each 'No' response across the five annotation features in Table B4, yielding a maximum possible score of 5.0. The highest scores are observed for How2Sign [18] (5.0), BOBSL [22] at 4.5, and RWTH-PHOENIX-Weather 2014 [23], SIGNUM [24], SWL-LSE [39], and LIBRAS-UFOP [43] (4.0 each).

4.7 Signer Demographics

Table B5 presents the full dataset-level detail of signer demographics. Demographic reporting across the 30 datasets is summarised in Table 7.

Table 7: Completeness of demographic reporting across the 30 included datasets

Demographic Field	Datasets Reporting	Not Reporting	% Complete
Number of Signers	30	0	100%
Deaf / Hearing / CODA Status	28	2	93%
Gender Distribution	14	16	47%
Age Range	11	19	37%

All 30 datasets report the number of signers, and 28 provide information on deaf, hearing, or CODA status. Gender distribution is reported by only 14 datasets, and age range by only 11. Among datasets reporting gender, distributions vary considerably: AUTSL [13] records 33 female and 10 male signers, while CASL-W60 [41] records 1 female and 18 male. Reported age ranges span from 6 to 18 years (Alabib-65 [37]) to 21 to 50 years (SIGNUM [24]). Complete signer-level demographic data, including CODA status and dialect variation, are presented in Table B5.

4.8 Accessibility and Availability

All 30 datasets are publicly available or accessible for research purposes. Distribution channels include GitHub, Zenodo, Kaggle, institutional repositories such as FIN-CLARIN, and access-on-request arrangements. Several datasets specify open licenses, such as Creative Commons Attribution 4.0 (YouTube-ASL [20]), CC BY-NC-ND 4.0

(OpenASL [21], KSU-SSL [36]), CC BY-NC-SA 4.0 (MM-WLAuslan [34]), and CC BY-NC (AzSLD [11], CASL-W60 [41]). Licensing terms are not specified in the source documentation for 15 datasets. Documentation quality varies: WLASL [9], AUTSL [13], BOBSL [22], and How2Sign [18] provide detailed collection descriptions, evaluation protocols, and download instructions, while several other datasets offer minimal guidance beyond their source publication.

4.9 Research Applications of the Mapped Datasets

This section outlines the research applications of the mapped datasets, specifying the tasks they support and documenting their use as benchmarks in the included publications. Many of these datasets serve as foundational references for evaluating model performance in isolated recognition, continuous recognition, and translation tasks. Table 8 consolidates these applications and lists the specific sources within this collection where each dataset is utilized or referenced.

Table 8: Research applications and cross-publication usage of the mapped datasets

Dataset	Task(s)	Sources Utilizing It
WLASL [9]	ISLR (Isolated word recognition)	Alabib-65 [37], MS-ASL [19], RoCoISLR [28], SignIT [40], MM-WLAuslan [34], OpenASL [21], Slovo [26], SWL-LSE [39], BdSLW60 [29], YouTube-ASL [20], CASL-W60 [41]
RWTH-PHOENIX-Weather 2014 [23]	CSLR, SLT	CSL-Daily [32], How2Sign [18], Mediapi-RGB [38], OpenASL [21], SignIT [40], YouTube-ASL [20], CE-CNSL [31], MS-ASL [19], BOBSL [22], RoCoISLR [28], Slovo [26], SMILE [33], KSL-Guide [42], AzSLD [11]
MS-ASL [19]	ISLR (Word-level classification)	AUTSL [13], Alabib-65 [37], SignIT [40], YouTube-ASL [20], BdSLW60 [29], MM-WLAuslan [34], RoCoISLR [28], Slovo [26], SWL-LSE [39], SL-ReDu GSL [10], BosphorusSign22k [25]
AUTSL [13]	ISLR (Multi-modal isolated recognition)	MM-WLAuslan [34], RoCoISLR [28], SignIT [40], Slovo [26], SWL-LSE [39], SL-ReDu GSL [10], BdSLW60 [29], BosphorusSign22k [25]

Dataset	Task(s)	Sources Utilizing It
How2Sign [18]	SLT (Instructional translation), Production	YouTube-ASL [20], Mediapi-RGB [38], OpenASL [21], RoCoISLR [28], AzSLD [11], KSL-Guide [42], SignIT [40]
CSL-Daily [32]	SLT, CSLR, ISLR (Isolated sign dictionary)	YouTube-ASL [20], OpenASL [21], CE-CNSL [31], BOBSL [22], AzSLD [11]
INCLUDE [30]	ISLR (Lexicon-scale recognition)	MM-WLAuslan [34], RoCoISLR [28], SignIT [40], SWL-LSE [39], BOBSL [22], CSL-Daily [32]
Bosphorus Sign22k [25]	ISLR (Chroma-key background research)	AUTSL [13], MM-WLAuslan [34], RoCoISLR [28], SWL-LSE [39], AzSLD [11]
SIGNUM [24]	CSLR (Signer-independent continuous recognition)	CE-CNSL [31], MS-ASL [19], BOBSL [22], SMILE [33], SL-ReDu GSL [10], LSA64 [35], How2Sign [18], SignIT [40]
SMILE [33]	L2 learner assessment (Automated feedback)	MS-ASL [19], BOBSL [22], BosphorusSign22k [25], MM-WLAuslan [34], SWL-LSE [39]
LSA64 [35]	ISLR (Outdoors and indoors word recognition)	WLASL [9], Alabib-65 [37], MM-WLAuslan [34], RoCoISLR [28], Slovo [26], SWL-LSE [39], BdSLW60 [29]
BOBSL [22]	SLR, Alignment, SLT, Sign-spotting	Mediapi-RGB [38], OpenASL [21], YouTube-ASL [20], AzSLD [11], SWL-LSE [39], SignIT [40]
YouTube-ASL [20]	SLT (Large-scale web mining), Alignment	AzSLD [11], CE-CNSL [31]
OpenASL [21]	SLT (Open-domain translation)	YouTube-ASL [20], AzSLD [11]
Slovo [26]	ISLR (Large-scale Russian benchmark)	AzSLD [11], SWL-LSE [39], BdSLW60 [29]
SL-ReDu GSL [10]	ISLR, CSLR, Fingerspelling, L2 Education	SWL-LSE [39], RoCoISLR [28]
Alabib-65 [37]	ISLR (Algerian subtle-sign recognition)	AzSLD [11]
BdSLW60 [29]	ISLR (Spatio-temporal word recognition)	CASL-W60 [41]

Dataset	Task(s)	Sources Utilizing It
MM-WLAuslan [34]	ISLR (Multi-view, multi-modal recognition)	RoCoISLR [28]
Mediapi-RGB [38]	SLT (Journalistic LSF translation), Sign Retrieval	SignIT [40], BOBSL [22]
KSU-SSL [36]	SLR (Medical and daily domain recognition)	BdSLW60 [29]
KSL-Guide [42]	SLT, CSLR, ISLR, SSLR (Interrogative sentences)	SignIT [40], YouTube-ASL [20], OpenASL [21]
AzSLD [11]	SLT, Word Recognition, Fingerspelling	YouTube-ASL [20], OpenASL [21]
CE-CNSL [31]	CSLR (Real-life scenario recognition)	CSL-Daily [32]
LIBRAS-UFOP [43]	ISLR (Minimal-pairs classification)	SignIT [40]
RoCoISLR [28]	ISLR (Romanian sign recognition)	—
CASL-W60 [41]	ISLR (Central African word classification)	—
SWL-LSE [39]	ISLR (Health-related Spanish signs)	—

4.9.1 Isolated Sign Language Recognition (ISLR)

WLASL [9] and MS-ASL [19] are the most frequently referenced benchmarks for word-level recognition, appearing in numerous cross-dataset studies. Reported performance results indicate significant challenges in scaling vocabulary: the best Inflated 3D Convolutional (I3D) model achieved 65.89% Top-1 accuracy on the WLASL100 subset, but only 32.48% on the full WLASL2000 subset [9]. Regional datasets provide essential ISLR baselines for under-represented languages. For example, BdSLW60 [29] for Bangla reports 97.0% accuracy using Support Vector Machine (SVM) on a flipped dataset, while AUTSL [13] for Turkish achieved 95.95% accuracy under random splits and 62.02% under signer-independent protocols. Specialized datasets such as LIBRAS-UFOP [43] (Brazilian Sign Language) and Alabib-65 [37] (Algerian Sign Language) focus on model precision using minimal-pairs vocabularies and subtle sign classes, reporting recognition rates of 74.25% and multi-method comparative baselines, respectively. Recent datasets such as RoCoISLR [28] (Romanian) and CASL-W60 [41] (Central African) utilize VideoMAE and transformer-based models to establish standardized evaluation foundations for their respective regions.

4.9.2 Continuous Sign Language Recognition (CSLR)

RWTH-PHOENIX-Weather 2014 [23] remains the primary benchmark for continuous sign language recognition, reporting a 45.0% Word Error Rate (WER) on single-signer setups as a baseline for upper-body tracking evaluation. SIGNUM [24] serves as a foundational benchmark for signer-independent continuous recognition, encompassing a vocabulary of 450 basic signs in German Sign Language. More recent datasets, such as CE-CNSL [31] (Chinese Sign Language), address the transition from laboratory settings to real-life scenarios, reporting an 18.6% WER using the TFNet time-frequency model. KSL-Guide [42] contributes by incorporating interrogative sentences into its training corpus; the authors report that this inclusion can nearly halve the WER for continuous recognition tasks in real-world application environments.

4.9.3 Sign Language Translation (SLT)

Large-scale multimodal datasets such as How2Sign [18] facilitate continuous ASL-to-English translation across more than 35,000 signed sentences and an English vocabulary of 16,000 words. CSL-Daily [32] introduced the Sign Back-Translation (SignBT) method for Chinese Sign Language translation, achieving a Bilingual Evaluation Understudy (BLEU) score of 21.34 by leveraging external monolingual data. BLEU measures the degree of overlap between machine-generated translations and human reference translations, with higher scores indicating closer agreement. Mediapi-RGB [38] offers the largest French Sign Language translation corpus, comprising 86 hours of journalistic content. Web-mined corpora such as YouTube-ASL [20] and OpenASL [21] significantly expand the scale of translation tasks. YouTube-ASL [20], for example, includes approximately 1,000 hours of video and more than 2,500 unique signers, and reports a state-of-the-art fine-tuned baseline of 12.39 BLEU on the How2Sign benchmark.

4.9.4 Specialized and Emerging Research Tasks

Several datasets address applications beyond general recognition tasks. BOBSL [22] supports large-scale sign language alignment and sign-spotting, reporting baseline results for subtitle alignment across diverse broadcast topics. Educational applications are central to SMILE [33] (Swiss German Sign Language) and SL-ReDu GSL [10] (Greek Sign Language), both designed for second-language (L2) learner assessment at the CEFR A1 level and which provide tools for student self-monitoring and automated feedback. Domain-specific communication is addressed by KSU-SSL [36], the largest Saudi Sign Language dataset in this collection, which focuses on healthcare communication to assist Deaf users in describing medical conditions. SWL-LSE [39] similarly targets health-related signs for Spanish Sign Language, reporting a mean accuracy of 91.83% using the MS-G3D baseline as part of the SignaMed online dictionary project.

5 Discussion

This chapter analyzes the results from Chapter 4 by identifying structural gaps and limitations across the 30 publicly available sign language datasets found through the systematic search. It proposes targeted recommendations for future dataset development and offers a task-based guide to help researchers select suitable datasets for specific computational tasks. All observations are based on the extracted data and are supported by prior literature, including Bragg et al. [1], Koller [2], Li et al. [9], Camgöz et al. [17], and Petersen et al. [6].

5.1 Identified Gaps and Limitations

The classification results indicate recurring gaps and limitations across the mapped datasets. These issues are organized thematically below, with the affected datasets identified by name in each sub-section.

5.1.1 Missing Signer-Independent Evaluation Protocols

Twelve datasets do not define a signer-independent evaluation split: CE-CNSL [31], Slovo [26], KSL-Guide [42], CSL-Daily [32], AzSLD [11], YouTube-ASL [20], Open-ASL [21], Corpus FinSL [27], RoCoISLR [28], Mediapi-RGB [38], KSU-SSL [36], and INCLUDE [30]. An additional two datasets, SL-ReDu GSL [10] and SignIT [40], do not report this information in their source publications. In the absence of a predefined signer-independent partition, evaluations may default to signer-dependent protocols, where the model is exposed to all test signers during training. The practical consequence of this is demonstrated by AUTSL, where signer-independent evaluation yielded 62.02% accuracy compared to 95.95% under a random split, in which samples from all signers are distributed arbitrarily across training and test sets without controlling for signer identity [13]. This gap of over 33 percentage points demonstrates that protocol choice can significantly affect reported performance. Petersen et al. [6] emphasize that the lack of standardized evaluation conditions is a structural barrier to meaningful cross-study comparison, a pattern evident in the current corpus.

5.1.2 Absent or Incomplete Sentence Alignment

Fifteen datasets lack sentence-level spoken-language translations, meaning they do not provide written equivalents that pair each signed sequence with a corresponding spoken-language sentence, which are essential for training translation models: LIBRAS-UFOP [43], BdSLW60 [29], BosphorusSign22k [25], AUTSL [13], WLASL [9], MS-ASL [19], Alabib-65 [37], MM-WLAuslan [34], CASL-W60 [41], SMILE [33], LSA64 [35], SignIT [40], RoCoISLR [28], INCLUDE [30], and KSU-SSL [36]. This limitation prevents their use in sign language translation research. The issue is particularly significant for larger datasets that offer the scale required for translation models but lack the necessary paired sentence annotations. For example, MS-ASL [19]

contains 222 signers and 25,513 video samples, AUTSL [13] has 43 signers and 38,336 samples, and MM-WLAuslan [34] includes 73 signers and over 282,900 samples, all without sentence-level translations. Camgöz et al. [17] demonstrated that paired gloss and sentence annotations directly enable more advanced translation architectures, which makes this gap a structural barrier to expanding translation research to under-resourced languages.

5.1.3 Insufficient Non-Manual Annotation

Non-manual features such as facial expressions, eyebrow movements, mouthing, and head and body posture convey grammatical and prosodic meaning in sign languages [1]. Twenty-one datasets lack non-manual annotation: MS-ASL [19], WLASL [9], YouTube-ASL [20], OpenASL [21], RWTH-PHOENIX-Weather 2014 [23], SIGNUM [24], AUTSL [13], Slovo [26], RoCoISLR [28], INCLUDE [30], CSL-Daily [32], MM-WLAuslan [9], LSA64 [35], AzSLD [11], KSU-SSL [36], Alabib-65 [37], SWL-LSE [39], CASL-W60 [41], KSL-Guide [42], Mediapi-RGB [38], and SMILE [33]. An additional six datasets provide only partial coverage: BOBSL [22], BosphorusSign22k [25], Corpus FinSL [27], CE-CNSL [31], BdSLW60 [29], and SignIT [40]. Only How2Sign [18], LIBRAS-UFOP [43], and SL-ReDu GSL [10] fully annotate these features. The widespread absence of non-manual annotation means that the majority of the corpus cannot support models that capture the full linguistic structure of sign languages, where non-manual features are essential for marking negation, questions, degree, and intensity.

5.1.4 Incomplete Demographic Reporting

Demographic information is reported inconsistently. Gender distribution is missing from 16 datasets: WLASL [9], MS-ASL [19], AUTSL [13], Slovo [26], YouTube-ASL [20], RoCoISLR [28], BdSLW60 [29], SignIT [40], KSL-Guide [42], AzSLD [11], BOBSL [22], RWTH-PHOENIX-Weather 2014 [23], INCLUDE [30], LSA64 [35], Mediapi-RGB [38], and KSU-SSL [36]. The age range is missing from 19 datasets, including all of the above, as well as MM-WLAuslan [34], OpenASL [21], How2Sign [18], KSL-Guide [42], and BosphorusSign22k [25]. Bragg et al. [1] show that the absence of demographic metadata makes it impossible to assess dataset representativeness or identify potential sources of bias in trained models. Without information on participants' ages, genders, and signing backgrounds, researchers cannot determine whether a dataset reflects the diversity of the community it is intended to represent.

5.1.5 Missing Duration Information

Eleven datasets do not report total duration: AUTSL [13], RoCoISLR [28], BdSLW60 [29], SMILE [33], MM-WLAuslan [34], LSA64 [35], Alabib-65 [37], SWL-LSE [39], CASL-W60 [41], KSL-Guide [42], and LIBRAS-UFOP [43]. Duration is a standard reporting requirement in video-based recognition benchmarks, as it determines the

volume of training data and is necessary for estimating computational cost. The absence of duration information in over one-third of the corpus limits direct scale comparisons and complicates researchers' ability to assess whether a dataset provides sufficient volume for a given task.

5.1.6 Very Small Signer Pools

Eight datasets include 12 or fewer signers: RWTH-PHOENIX-Weather 2014 [23] (9), How2Sign [18] (11), Mediapi-RGB [38] (10), CSL-Daily [32] (10), CE-CNSL [31] (12), INCLUDE [30] (7), BosphorusSign22k [25] (6), and LIBRAS-UFOP [43] (5). Small signer pools limit the diversity of signing styles represented. Li et al. [9] show that limited signer diversity can artificially inflate accuracy under signer-dependent evaluation, as the model is exposed to all signing styles during training. This issue is relevant even for RWTH-PHOENIX-Weather 2014 [23], the most widely used CSLR benchmark, which includes only 9 signers.

5.1.7 Geographic and Linguistic Underrepresentation

This study identified only two datasets from Africa and two from Latin America through its systematic search. ASL accounts for five of the 30 identified datasets and DGS for two, indicating a concentration of publicly available resources around a small number of sign languages. No dataset was included for sign languages from South-East Asia, Central Asia, the Middle East outside Saudi Arabia, or Sub-Saharan Africa beyond the two identified, as the initial keyword search yielded sparse results. Bragg et al. [1] highlight geographic and linguistic concentration as a structural limitation in sign language AI research, noting that many deaf communities worldwide lack computational resources. The distribution observed in this mapping study aligns with this finding. Even within better-represented regions, the distribution remains unequal: the five ASL datasets outnumber the combined total from Africa and Latin America. The lack of datasets for sign languages such as Indonesian, Japanese, Nigerian, and Pakistani Sign Language in the mapped corpus highlights the structural inequality in research investment described by Bragg et al. [1]. The systematic search conducted in this study did not identify publicly available resources for these languages.

5.2 Recommendations for Improving Sign Language Datasets

Based on the gaps identified in Section 5.1 and supported by prior literature, the following recommendations are provided for researchers and institutions developing or extending sign language datasets.

5.2.1 Define and Publish Signer-Independent Evaluation Splits

All datasets intended for benchmarking should define and release a standardized signer-independent evaluation split alongside the data. The AUTSL results illustrate this need: signer-independent evaluation produced results more than 33 percentage points

lower than the random-split result, yet the signer-independent figure more accurately reflects real-world performance on unseen individuals [13]. Publishing a fixed split, a predetermined and publicly released partition of the data used identically by all researchers, ensures that results from different research groups are directly comparable. Petersen et al. [6] identify this as a prerequisite for meaningful benchmarking.

5.2.2 Add Sentence-Level Translations to Continuous Datasets

Continuous datasets that lack spoken-language translations should include supplementary annotation layers that pair signed segments with written sentences. The utility of such annotation is demonstrated by RWTH-PHOENIX-Weather 2014 [23], where paired gloss and sentence annotations, aligning each signed sequence with both a gloss-level transcription and a full spoken-language sentence, enabled transformer-based joint recognition and translation models [17]. Large-signer-pool datasets such as AUTSL [13], MS-ASL [19], and MM-WLAuslan [34] are particularly strong candidates for annotation extension due to their scale.

5.2.3 Report and Standardize Demographic Metadata

All dataset documentation should consistently report the number of signers, deaf or hearing, or CODA status, gender distribution, and age range. These four fields should be considered mandatory minimum requirements. Bragg et al. [1] show that the absence of demographic information prevents assessment of representativeness and potential bias. Datasets should also indicate whether signers are native language users or learners, as this distinction directly affects the linguistic quality of the recorded data.

5.2.4 Incorporate Non-Manual Feature Annotations

Future datasets should include annotation of non-manual features such as facial expressions, eyebrow position, mouthing, and head movements, which convey grammatical meaning in sign languages. As only 3 of the 30 mapped datasets fully annotate these features, there is a significant opportunity for new releases to address this gap. Tools such as ELAN and OpenFace can facilitate this process, and retrospective annotation of existing datasets is technically feasible.

5.2.5 Report Total Duration Consistently

All dataset documentation should report total duration in hours or minutes, calculated from frame counts and frame rates if direct measurement is unavailable. This practice is standard in comparable video recognition benchmarks and should be considered a minimum requirement for sign language datasets.

5.2.6 Expand Geographic and Linguistic Coverage

Future dataset development should prioritize sign languages currently absent from the research literature, including those from Sub-Saharan Africa, South-East Asia, Central Asia, and the Middle East beyond Saudi Arabia. Coordinated international initiatives in close collaboration with local deaf communities are most likely to yield representative and linguistically accurate datasets. Bragg et al. [1] specifically recommend community-centered approaches to dataset design to address the structural inequalities evident in the current distribution.

5.3 Task-Based Dataset Usage Guidelines

This section offers a structured guide for dataset selection based on the computational task a researcher intends to pursue. Recommendations are based on annotation, modality, and signer pool characteristics recorded in the extraction spreadsheet. Table 9 presents a concise overview, and the following sub-sections discuss each task type in detail.

5.3.1 Isolated Sign Language Recognition

For isolated sign language recognition (ISLR), researchers should select datasets that provide gloss labels and a published signer-independent split. WLASL [9] and MS-ASL [19] are recommended for large-vocabulary ASL recognition. AUTSL [13] and BosphorusSign22k [25] are suitable for Turkish Sign Language with RGB and depth modalities. MM-WLAuslan [34] is the preferred choice for Australian Sign Language, offering over 282,900 samples with a multi-view configuration. Slovo [26] provides the most subject-diverse Russian Sign Language resource, with 194 signers across 1,000 classes. For under-resourced languages, INCLUDE [30], Alabib-65 [37], RoCoISLR [28], SignIT [40], and CASL-W60 [41] are the only available benchmarks for Indian, Algerian, Romanian, Italian, and Central African Sign Languages, respectively. It should be noted that KSU-SSL [36], INCLUDE [30], and RoCoISLR [28] do not define signer-independent splits, which limits the rigor of evaluations on these datasets.

5.3.2 Continuous Sign Language Recognition

Continuous sign language recognition (CSLR) requires temporal gloss annotations, time-stamped labels marking the start and end boundaries of each sign within a continuous video sequence, and a signer-independent split. RWTH-PHOENIX-Weather 2014 [23] remains the most widely used CSLR benchmark, with 1,558 glosses, 9 signers, a standardized split, and paired sentence translations. SIGNUM [24] provides a controlled German Sign Language benchmark with 450 signs from 25 native signers. BOBSL [22] supports large-scale British Sign Language recognition with approximately 1,467 hours of broadcast footage. CE-CNSL [31] is recommended for recognition under natural, complex environmental conditions, given its more than 70 distinct background settings.

Table 9: Task-based dataset selection guide. Recommendations are based on annotation features, split availability, modality, and signer pool size as extracted from the source publications

Research Task	Key Requirements	Recommended Datasets
Isolated SLR (word-level)	Gloss labels; signer-independent split; large vocabulary	WLASL [9], MS-ASL [19], AUTSL [13], MM-WLAuslan [34], Slovo [26], INCLUDE [30], RoCoISLR [28], BosphorusSign22k [25], SWL-LSE [39], Alabib-65 [37], CASL-W60 [41], LSA64 [35]
Continuous SLR	Temporal gloss annotations; signer-independent split	RWTH-PHOENIX-Weather 2014 [23], SIGNUM [24], BOBSL [22], CE-CNSL [31]
Sign Language Translation	Sentence alignment, large vocabulary, sufficient signer pool	How2Sign [18], CSL-Daily [32], YouTube-ASL [20], OpenASL [21], Mediapi-RGB [38], KSL-Guide [42], RWTH-PHOENIX-Weather 2014 [23], Corpus FinSL [27]
Sign Spotting	Continuous video; temporal gloss annotations; multiple signers	BOBSL [22], SL-ReDu GSL [10]
Fingerspelling Recognition	Letter-level annotations; fingerspelling subset	AzSLD [11], SL-ReDu GSL [10], KSL-Guide [42]
Multi-modal / Depth-based SLR	RGB + Depth or Skeleton modalities	AUTSL [13], BosphorusSign22k [25], SMILE [33], MM-WLAuslan [34], LIBRAS-UFOP [43], SL-ReDu GSL [10], How2Sign [18]
Multi-view SLR	Multi-camera setup; synchronized angles	MM-WLAuslan [34], How2Sign [18], SMILE [33], AzSLD [11], KSL-Guide [42]
L2 Learner Assessment	Native and learner signer pairs; fine-grained annotation	SMILE [33]
Domain-Specific SLR	Domain vocabulary; controlled conditions	RWTH-PHOENIX-Weather 2014 (weather) [23], KSU-SSL (10 domains) [36], SWL-LSE (health) [39], SL-ReDu GSL (education) [10]
Low-Resource SL Research	Under-represented sign language; any modality	Alabib-65 [37], CASL-W60 [41], LIBRAS-UFOP [43], INCLUDE [30], RoCoISLR [28], SignIT [40], SMILE [33]

5.3.3 Sign Language Translation

Sign language translation (SLT) requires sentence-level translations paired with signed video, a large vocabulary, and a sufficient signer pool. RWTH-PHOENIX-Weather 2014 [23] is the most established SLT benchmark. For American Sign Language, How2Sign [18], YouTube-ASL [20], and OpenASL [21] are recommended; OpenASL [21] is the largest, with 288 hours and over 33,000 vocabulary items. CSL-Daily [32] supports Chinese Sign Language translation. Mediapi-RGB [38] is recommended for French Sign Language, providing 86 hours produced by deaf journalists. KSL-Guide [42] supports Korean Sign Language translation and includes a unique subset of interrogative sentences.

5.3.4 Sign Spotting, Fingerspelling, and Specialist Tasks

Sign-spotting research has a limited dataset support; BOBSL [22] and SL-ReDu GSL [10] are the primary resources. For fingerspelling, AzSLD [11] includes 32 letter-level fingerspelling classes for Azerbaijani Sign Language, and KSL-Guide [42] includes fingerspelled samples for Korean Sign Language. Multi-modal research requiring depth data should utilize AUTSL [13], BosphorusSign22k [25], SMILE [33], MM-WLAuslan [34], LIBRAS-UFOP [43], SL-ReDu GSL [10], or How2Sign [18]. For multi-view research, MM-WLAuslan [34], How2Sign [18], SMILE [33], AzSLD [11], and KSL-Guide [42] are suitable. SMILE [33] is the only dataset in the corpus designed for automated assessment of L2 learners.

5.4 Summary

The findings of this systematic mapping study reveal both the breadth and the structural limitations of the 30 publicly available sign language datasets identified through the systematic search. Collectively, these datasets cover 23 sign languages across five geographic regions and support a range of computational tasks, from isolated-word recognition to large-scale translation. However, the corpus is uneven, dominated by isolated datasets, concentrated in a small number of languages and regions, and characterized by inconsistent annotation practices and incomplete demographic reporting.

The most pervasive limitations are the absence of signer-independent evaluation splits in 12 datasets, sentence-level translations in 15 datasets, and non-manual annotations in 21 datasets. These three gaps collectively prevent a substantial portion of the corpus from supporting research tasks such as translation and linguistic modeling, which represent the current frontier of sign language AI research. The recommendations in Section 5.2 and the usage guidelines in Section 5.3 are designed to address these limitations in practical terms. Progress will depend less on collecting additional data of the existing kind and more on improving annotation depth, evaluation rigor, and documentation consistency in current resources, alongside targeted investment in underrepresented sign languages.

6 Conclusion

This thesis employed the systematic mapping methodology of Petersen et al. [6, 7] and adhered to the guidelines of Kitchenham and Charters [8] to generate a structured overview of publicly available sign language datasets. Thirty datasets were identified through a keyword-based search across five academic databases, supplemented by backward snowballing. These datasets were analyzed using a predefined 38-attribute extraction scheme, organized into six classification dimensions: language coverage and geographic distribution, dataset type and supported tasks, scale and composition, modality and recording conditions, annotation and benchmarking practices, and signer demographics and accessibility. The complete data extraction sheets are openly accessible via the project repository linked in Appendix A, to support replication and extension of this study. The three research sub-questions formulated in Chapter 3 are addressed as follows:

1. Regarding the structural and technical characteristics of the existing datasets, the mapped corpus encompasses 23 distinct sign languages across five geographic regions, with a notable concentration in Europe and the Asia-Pacific. RGB video is nearly universal, present in 29 of the 30 datasets, whereas richer modalities such as depth video and multi-view camera configurations are available in only 7 and 5 datasets, respectively. Annotation depth varies considerably: gloss labels and baseline results are reported by 90 percent of datasets, while non-manual annotation is fully present in only 3 datasets, and sentence-level alignment in only 15.

2. In terms of supported computational tasks, isolated sign language recognition is the most strongly supported, with 16 datasets providing the necessary gloss labels and signer-independent splits. Continuous sign language recognition and sign language translation are less widely supported, with only 9 continuous datasets and 15 datasets providing the sentence alignment required for translation tasks. Sign-spotting and fingerspelling recognition each depend on a small number of specialized resources.

3. The most structurally significant limitations are the absence of signer-independent evaluation splits in 12 datasets, missing sentence-level translations in 15 datasets, and the near-complete absence of non-manual annotation across the corpus.

The principal contribution of this thesis is a systematic, evidence-based characterization of the sign language dataset landscape, which focuses on datasets as the primary object of analysis rather than the models trained on them. Previous surveys, such as those by Koller [2], Sarhan and Frintrop [4], and Tan et al. [14], employ datasets as benchmarks for evaluating recognition and translation systems. Structured review studies, such as Wadhawan and Kumar [5], classify methods and algorithms rather than dataset characteristics. The ethics-oriented analysis by Bragg et al. [1] critically examines datasets but does not employ a systematic mapping framework. This thesis addresses the gap left by these prior contributions by providing a structured, replicable

classification of dataset characteristics aligned with predefined research questions, consistent with the requirements of Petersen et al. [6]. The practical implications of this study are directed toward two primary audiences. For researchers selecting datasets for specific tasks, the task-based usage guidelines in Section 5.3 and the selection table in Table 9 offer evidence-based starting points organized by annotation requirements, modality, and signer pool size. For dataset developers and funding bodies, the six recommendations in Section 5.2 highlight the most consequential areas for improvement: publishing signer-independent evaluation splits, adding sentence-level translations to large-scale datasets, standardizing demographic reporting, incorporating non-manual feature annotation, reporting total duration consistently, and expanding geographic and linguistic coverage.

Three directions warrant attention in future work. First, the mapping should be updated as new datasets are published, given that several datasets in this corpus date from 2024 and 2025, and the field is evolving rapidly. Second, large-signer-pool datasets that currently lack sentence-level annotation, such as MS-ASL [19] and AUTSL [13], are strong candidates for retrospective annotation extension, as their scale already meets the requirements for translation research. Third, targeted dataset collection for sign languages from Sub-Saharan Africa, East and South-East Asia, and Central Asia remains the most structurally significant unmet need identified by this study, as no publicly available datasets were identified for sign languages such as Japanese, Indonesian, and Pakistani Sign Language through the systematic search.

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A Appendix: Data Extraction Sheets

A.1 Data Extraction Sheet

Data extraction sheet can be found here: https://github.com/knzkoly/thesis-systematic-mapping/blob/main/Dataset_Mapping.xlsx

A.2 Standardized Data Extraction Sheet

Standardized data extraction sheet can be found here: https://github.com/knzkoly/thesis-systematic-mapping/blob/main/Dataset_Mapping_Cleaned.xlsx

B Appendix: Thematic Attribute Tables

B.1 Scale and Scope

Table B1 presents the scale and scope of all 30 included datasets, sorted in extraction order. The columns report the year of publication, dataset type, number of sign classes or vocabulary items, number of video samples, total duration in hours, and number of signers. A dash (—) indicates that the value was not specified in the source publication. The #Sign Classes column reports the number of distinct sign glosses or word classes in each dataset. The #Video Samples column reports the total number of video clips or sequences. Total Hours reflects the cumulative duration of video footage as reported by the original authors. #Signers reports the total number of individual signing participants.

Table B1: Scale and scope of the 30 sign language datasets included in the systematic mapping study

Dataset	Year	Type	#Sign Classes	#Video Samples	Total Hours	#Signers
How2Sign	2021	Continuous	16,000 words	2,653	82.08 h	11
MS-ASL	2019	Isolated	1,000	25,513	~24.0 h	222
WLASL	2020	Isolated	2,000	21,083	~14.0 h	119
YouTube-ASL	2023	Continuous	60,000	11,093	984.0 h	2,519
OpenASL	2022	Continuous	33,549	98,417	288.0 h	220
BOBSL	2021	Continuous	2,281	1,962	~1,467 h	39
RWTH-PHOENIX-Weather 2014	2014	Continuous	1,558	645	10.73 h	9
SIGNUM Database	2010	Mixed	450	33,210	55.3 h	25
AUTSL	2020	Isolated	226	38,336	—	43
Bosphorus Sign22k	2020	Isolated	744	22,542	~19.0 h	6
Slovo	2023	Isolated	1,000	20,000	19.81 h	194
Corpus FinSL	2020	Continuous	107,000 tokens	343	14.37 h	21
RoCoISLR	2025	Isolated	5,892	9,141	—	25
BdSLW60	2025	Isolated	60	9,307	—	18
INCLUDE	2020	Isolated	263	4,287	~3.0 h	7
CE-CNSL	2024	Continuous	3,515	5,988	10.52 h	12
CSL-Daily	2021	Continuous	2,000	20,654	23.27 h	10

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Dataset	Year	Type	#Sign Classes	#Video Samples	Total Hours	#Signers
SL-ReDu GSL corpus	2023	Mixed	369	30,116	35.93 h	21
SMILE	2018	Isolated	100	9,000	—	30
MM-WLAuslan	2024	Isolated	3,215	282,900	—	73
LSA64	2023	Isolated	64	3,200	—	10
AzSLD	2025	Mixed	719	30,312	65.2 h	43
KSU-SSL	2023	Mixed	293	145,035	495.0 h	33
Alabib-65	2023	Isolated	65	6,238	—	41
Mediapi-RGB	2024	Continuous	35,599 subtitles	1,230	86.0 h	>10
SWL-LSE	2024	Isolated	300	8,000	—	124
SignIT	2025	Isolated	94	644	3.33 h	45
CASL-W60	2025	Isolated	60	5,889	—	19
KSL-Guide	2021	Mixed	2,000	521,000	—	72
LIBRAS-UFOP	2021	Isolated	56	3,040	—	5

B.2 Modality Coverage

Table B2 presents modality coverage for all 30 datasets. "Yes" indicates that the modality is included in the released dataset. "No" indicates that the modality is explicitly absent. "Partial" indicates that the modality is present for a subset of the data or is derived rather than directly captured. Skeleton / Pose data refers to 2D or 3D body, hand, and face keypoints extracted using tools such as OpenPose, MediaPipe, or AlphaPose, or captured via motion-capture equipment.

Table B2: Modality coverage across the 30 included datasets

Dataset	RGB Video	Depth Video	Skeleton / Pose	Multi-view	Audio
How2Sign	Yes	Yes	Yes	Yes	Yes
MS-ASL	Yes	No	Yes	No	No
WLASL	Yes	No	Yes	No	No
YouTube-ASL	Yes	No	No	No	No
OpenASL	Yes	No	No	No	No
BOBSL	Yes	No	Yes	No	No
RWTH-PHOENIX-Weather 2014	Yes	No	No	No	Yes
SIGNUM Database	Yes	No	No	No	No
AUTSL	Yes	Yes	Yes	No	No
BosphorusSign22k	Yes	Yes	Yes	No	No

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Dataset	RGB Video	Depth Video	Skeleton / Pose	Multi-view	Audio
Slovo	Yes	No	No	No	No
Corpus FinSL	Yes	No	No	No	No
RoCoISLR	Yes	No	No	No	No
BdSLW60	Yes	No	Yes	Partial	No
INCLUDE	Yes	No	Yes	No	No
CE-CNSL	Yes	No	No	No	No
CSL-Daily	Yes	No	No	No	No
SL-ReDu GSL corpus	Yes	Yes	No	No	No
SMILE	Yes	Yes	Yes	Yes	No
MM-WLAuslan	Yes	Yes	Yes	Yes	No
LSA64	Yes	No	Yes	No	No
AzSLD	Yes	No	No	Yes	No
KSU-SSL	Yes	No	Yes	No	No
Alabib-65	Yes	No	No	No	No
Mediapi-RGB	Yes	No	No	No	Partial
SWL-LSE	Partial	No	Yes	No	No
SignIT	Yes	No	Yes	No	No
CASL-W60	Yes	No	Yes	No	No
KSL-Guide	Yes	No	Yes	Yes	No
LIBRAS-UFOP	Yes	Yes	Yes	No	No

B.3 Recording Conditions

Table B3 presents recording conditions for all 30 datasets. The column headers are abbreviated as follows: Env. refers to recording environment, BG refers to background complexity, and FPS refers to frames per second. Environment classifies the recording setting as Studio (controlled laboratory or studio setup), Natural (real-world or in-the-wild recording), Broadcast (TV or online broadcast footage), Online (footage sourced from online platforms such as YouTube), Mixed (combination of conditions), or Lab (controlled laboratory environment). Background and Lighting classifications are derived from descriptions in the source publications. A dash (—) indicates that the information was not specified.

Table B3: Recording conditions of the 30 included datasets

Dataset	Env.	BG	Lighting	Resolution	FPS
How2Sign	Studio	Simple	—	1280 × 720	30
MS-ASL	Natural	Complex	Varied	Standardised	Varied
WLASL	Online	Simple	Varied	Varied	Varied

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Dataset	Env.	BG	Lighting	Resolution	FPS
YouTube-ASL	Online	Real-world	Varied	480 × 360	Varied
OpenASL	Online	Real-world	Varied	224 × 224	25
BOBSL	Broadcast	Simple	Controlled	444 × 444	25
RWTH-PHOENIX-Weather 2014	Studio	Controlled	Controlled	210 × 260	25
SIGNUM Database	Lab	Simple	Controlled	776 × 578	30
AUTSL	Mixed	Complex	Varied	512 × 512	—
Bosphorus-Sign22k	Studio	Simple	—	RGB: 1920×1080; Depth: 512×424	30
Slovo	Natural	Varied	Varied	1280 × 720	30
Corpus FinSL	Studio	—	—	—	—
RoCoISLR	Mixed	Simple	—	224 × 224	25
BdSLW60	Natural	Complex	—	—	Varied
INCLUDE	Natural	Complex	Natural	1920 × 1080	25
CE-CNSL	Natural	Complex	Natural	Varied	Varied
CSL-Daily	Lab	—	—	1920 × 1080	30
SL-ReDu GSL corpus	Studio	Simple	Controlled	1920 × 1080	RGB: 60Hz; Depth: 30Hz
SMILE	Studio	Simple	Controlled	Varied	Varied
MM-WLAuslan	Studio	Simple	Controlled	—	—
LSA64	Mixed	Simple	Mixed	—	—
AzSLD	Mixed	Simple	Controlled	224 × 224	30
KSU-SSL	Studio	Simple	Controlled	1920 × 1200	60
Alabib-65	Mixed	Complex	Varied	Varied	30
Mediapi-RGB	Natural	Complex	Professional	Varied	Varied
SWL-LSE	Natural	Complex	Varied	Varied	Varied
SignIT	Natural	Complex	Varied	Varied	Varied
CASL-W60	Mixed	Complex	Varied	224 × 224	30
KSL-Guide	Mixed	Mixed	Varied	1920 × 1080	30

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Dataset	Env.	BG	Lighting	Resolution	FPS
LIBRAS-UFOP	Mixed	Mixed	Controlled	—	30

B.4 Annotation and Benchmarking Characteristics

Table B4 presents annotation and benchmarking characteristics for all 30 datasets. A dash (—) indicates that the information was not reported in the source publication. Gloss Labels indicate whether the dataset provides word-level sign annotations aligned with video segments. Sentence Alignment indicates whether signed utterances are aligned with spoken or written language translations. Non-Manual Annotation indicates whether facial expressions, mouthing, or other non-manual features are explicitly annotated. Signer-Independent Split indicates whether the evaluation protocol excludes training-set signers from the test set. Baseline results indicate whether the source publication reports model performance results on the dataset.

Table B4: Annotation and benchmarking characteristics of the 30 included datasets

Dataset	Gloss Labels	Sentence Alignment	Non-Manual Annot.	Signer-Indep. Split	Baseline Results
How2Sign	Yes	Yes	Yes	Yes	Yes
MS-ASL	Yes	No	No	Yes	Yes
WLASL	Yes	No	No	Yes	Yes
YouTube-ASL	No	Yes	No	No	Yes
OpenASL	No	Yes	No	No	Yes
BOBSL	Yes	Yes	Partial	Yes	Yes
RWTH-PHOENIX-Weather 2014	Yes	Yes	No	Yes	Yes
SIGNUM Database	Yes	Yes	No	Yes	Yes
AUTSL	Yes	No	No	Yes	Yes
Bosphorus-Sign22k	Yes	No	Partial	Yes	Yes
Slovo	Yes	Yes	No	No	Yes
Corpus FinSL	Yes	Yes	Partial	No	No
RoCoISLR	Yes	No	No	No	Yes
BdSLW60	Yes	No	Partial	Yes	Yes
INCLUDE	Yes	No	No	No	Yes
CE-CNSL	Yes	Yes	Partial	No	Yes
CSL-Daily	Yes	Yes	No	No	Yes
SL-ReDu GSL corpus	Yes	Yes	Yes	—	No
SMILE	Yes	No	No	Yes	No
MM-WLAuslan	Yes	No	No	Yes	Yes

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Dataset	Gloss Labels	Sentence Align-ment	Non-Manual Annot.	Signer-Indep. Split	Baseline Results
LSA64	Yes	No	No	Yes	Yes
AzSLD	Yes	Yes	No	No	Yes
KSU-SSL	Yes	No	No	No	Yes
Alabib-65	Yes	No	No	Yes	Yes
Mediapi-RGB	No	Yes	No	No	Yes
SWL-LSE	Yes	Yes	No	Yes	Yes
SignIT	Yes	No	Partial	—	Yes
CASL-W60	Yes	No	No	Yes	Yes
KSL-Guide	Yes	Yes	No	No	Yes
LIBRAS-UFOP	Yes	No	Yes	Yes	Yes

B.5 Signer Demographics

Table B5 presents signer demographic information reported across the 30 datasets. The #Signers column reports the total number of signing participants as stated by the authors. Deaf / Hearing / CODA reports the signer community composition, where disclosed; CODA refers to Children of Deaf Adults. Gender Distribution is reported as F (female) and M (male), where provided. Age Range is extracted from the source text. A dash (—) indicates that the information was not reported.

Table B5: Signer demographics reported across the 30 included datasets

Dataset	#Signers	Deaf / Hearing / CODA	Gender Distribution	Age Range
How2Sign	11	4 Deaf, 5 Hearing, 2 Hard-of-hearing	5F / 6M	—
MS-ASL	222	Students / Teachers	—	—
WLASL	119	Native / Expert	—	—
YouTube-ASL	>2,519	All skill levels	—	—
OpenASL	>220	Native	48.2% F / 42.7% M / 9.1% other	Child – older adult
BOBSL	39	Hearing and deaf interpreters	—	—
RWTH-PHOENIX-Weather 2014	9	Hearing interpreters	—	—
SIGNUM Database	25	23 Deaf, 2 Hearing-impaired	13F / 12M	21–50

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Dataset	#Signers	Deaf / Hearing / CODA	Gender Distribution	Age Range
AUTSL	43	1 Deaf, 1 CODA, 6 TSL instructors, 3 TSL translators, 25 TSL course students, 7 trained signers	33F / 10M	—
Bosphorus-Sign22k	6	Native (Deaf)	—	—
Slovo	194	Crowdworkers	—	—
Corpus FinSL	21	Native (Deaf)	—	—
RoCoISLR	25	Hearing / Mixed	—	—
BdSLW60	18	Hearing volunteers	—	—
INCLUDE	7	Deaf students	—	—
CE-CNSL	12	2 Hearing-impaired, 10 Interpreters	8F / 4M	20–45
CSL-Daily	10	Native (Deaf community)	—	—
SL-ReDu GSL corpus	21	Deaf (L1), Bilingual	11F / 10M	19–52
SMILE	30	L1 signers + L2 learners	—	Adults
MM-WLAuslan	73	Deaf, Experts, Volunteers	Volunteers: 26.0% F / 39.7% M; Deaf: 12.3% F / 8.2% M	—
LSA64	10	Non-expert subjects	—	—
AzSLD	43	Native, Interpreters, Users	Diverse	—
KSU-SSL	33	Deaf, Hard-of-hearing, Hearing volunteers	—	—
Alabib-65	41	38 Deaf, 3 Hard-of-hearing	20F / 21M	6–18
Mediapi-RGB	>10	Deaf journalists	—	—
SWL-LSE	124	Deaf, Interpreters, LSE Learners	—	—
SignIT	45	—	—	—
CASL-W60	19	Trained volunteers	1F / 18M	22–35
KSL-Guide	72	Native KSL signers	35F / 17M	—
LIBRAS-UFOP	5	Trained volunteers	3F / 2M	—